

Single Column Model (SCM) Diagnostic Runs

GSPT Simulation Pipeline

July 15, 2026

1 Overview

This document compiles every current SCM case report under results/ that has a compiled wrapper PDF.

2 Case 1: idealized sbl 100x2m 9h report

3 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/bistable_dt10/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.))` and returned `Success`.

3.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/bistable_dt10`
- Number of saved time samples: 3241
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 407 / 4
- RHS evaluations: 126611
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

3.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.008131, 46.871441]$

3.3 Generated Artifacts

- Time-series CSV: `results/bistable_dt10/time_series.csv`
- Diagnostic summary JSON: `results/bistable_dt10/summary.json`
- Payload JLD2: `results/bistable_dt10/payload.jld2`

3.4 Figure Manifest

1. Time series of `T_s`, sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of `U`, `theta`, and `Km` at representative times
5. Surface energy budget components (`R_n`, `H`, `G`, storage)
6. Phase portrait on regularized manifold (`Delta` vs `e_xi`)
7. Eddy diffusivity response (`Km`, `Kh`) versus stability/Richardson number
8. Fold proximity diagnostics versus time

3.5 Included Plots

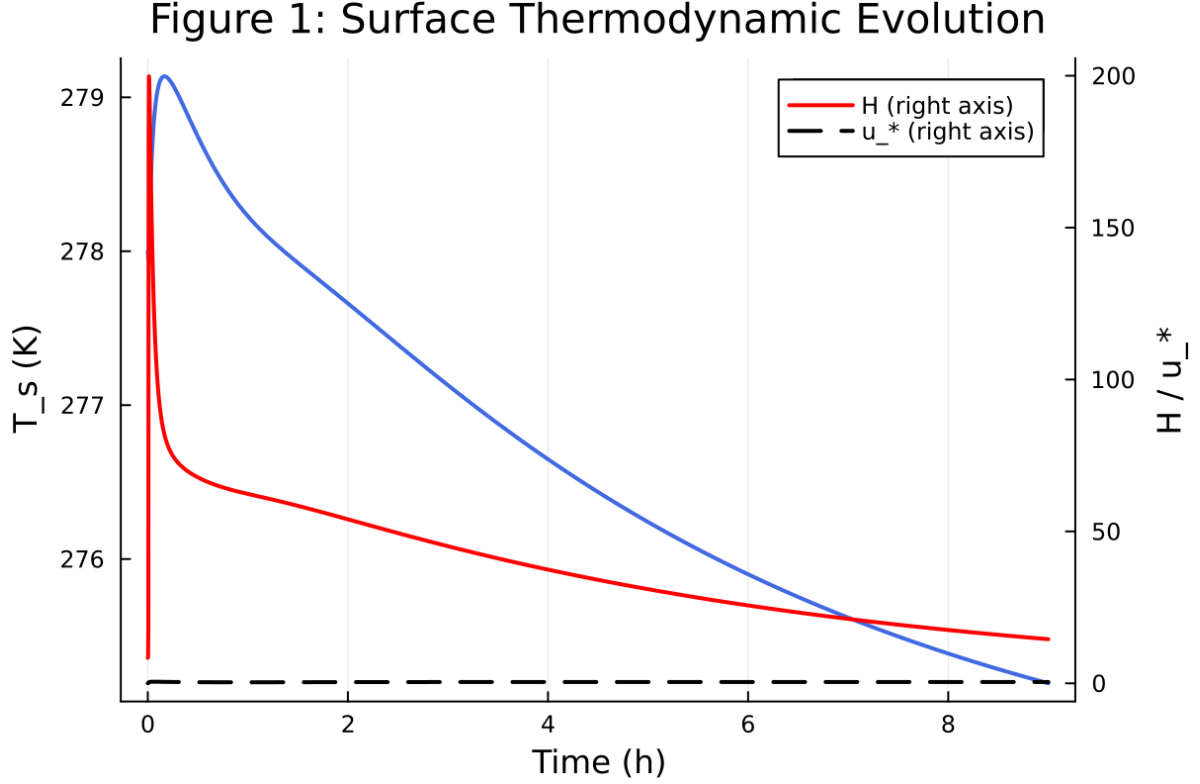


Figure 1: Time series of T_s , sensible heat flux, and friction velocity

4 Case 2: idealized sbl 100x2m 9h report

5 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/bistable_dt60/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver{Val{:forward}}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.tolerances{abstol=1.000e-08, reltol=1.000e-06})}` and returned `Success`.

5.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/bistable_dt60`
- Number of saved time samples: 541
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 407 / 4
- RHS evaluations: 126611
- Solver tolerances: `abstol = 1.000 × 10-8, reltol = 1.000 × 10-6`

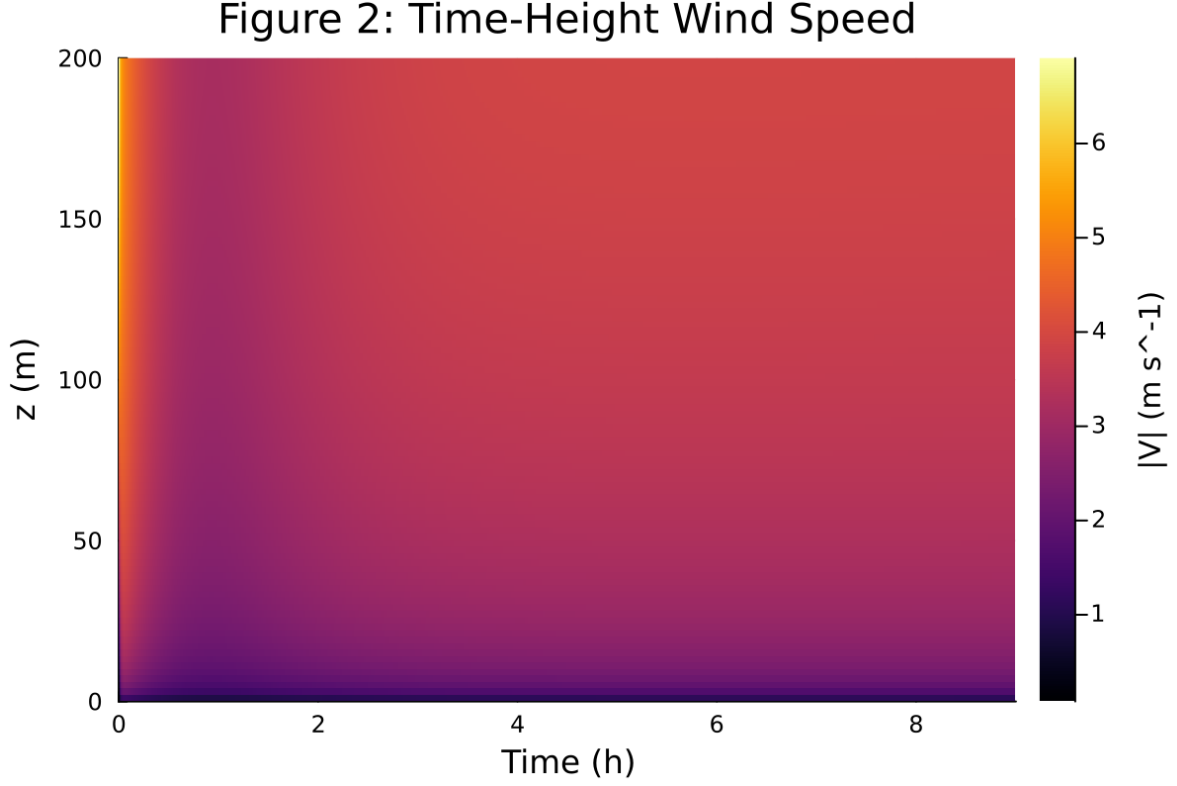


Figure 2: Time-height contour of wind speed with LLJ evolution

5.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.008131, 46.085434]$

5.3 Generated Artifacts

- Time-series CSV: `results/bistable_dt60/time_series.csv`
- Diagnostic summary JSON: `results/bistable_dt60/summary.json`
- Payload JLD2: `results/bistable_dt60/payload.jld2`

5.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

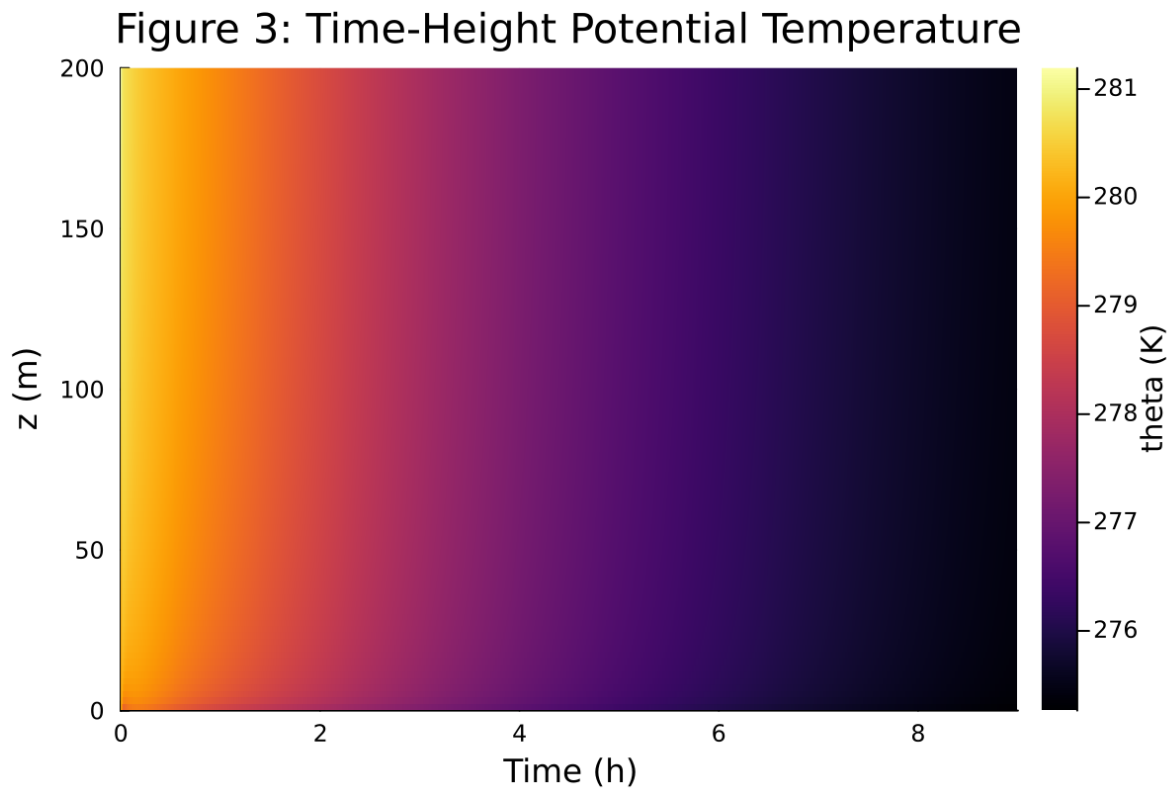


Figure 3: Time-height contour of potential temperature and inversion growth

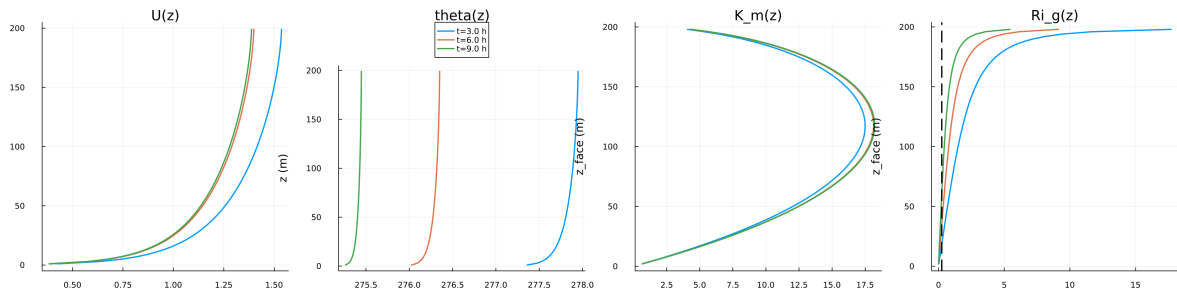


Figure 4: Vertical profiles of U , θ , and K_m at representative times

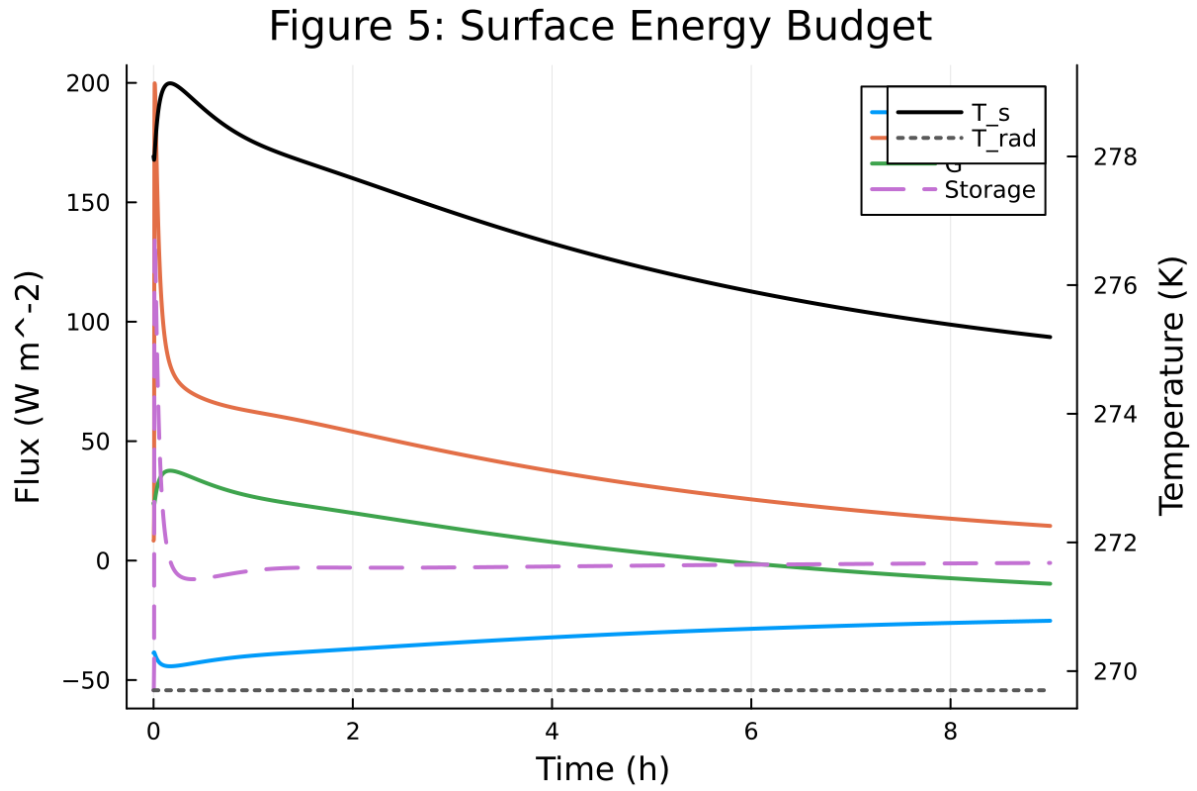


Figure 5: Surface energy budget components (R_n , H , G , storage)

6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

5.5 Included Plots

Figure 6: Regularized Slow-Manifold Phase Portrait

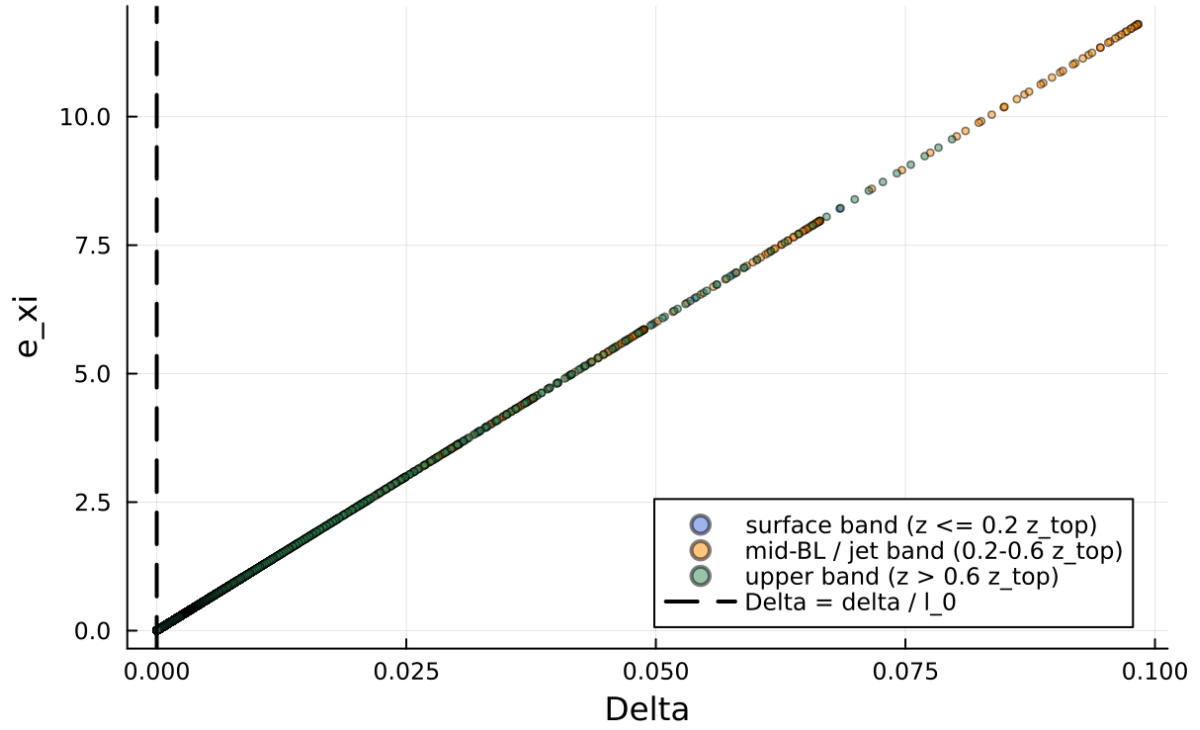


Figure 6: Phase portrait on regularized manifold (Δ vs e_{xi})

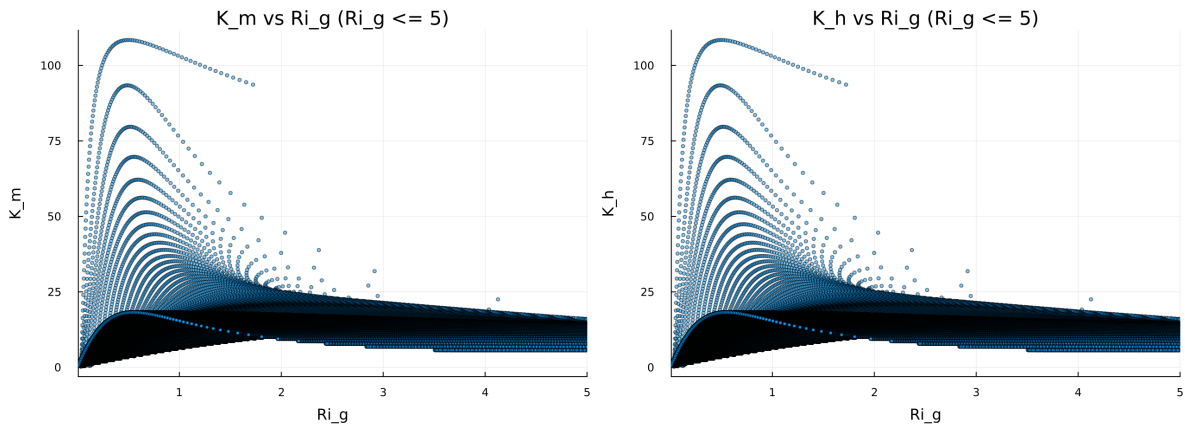


Figure 7: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

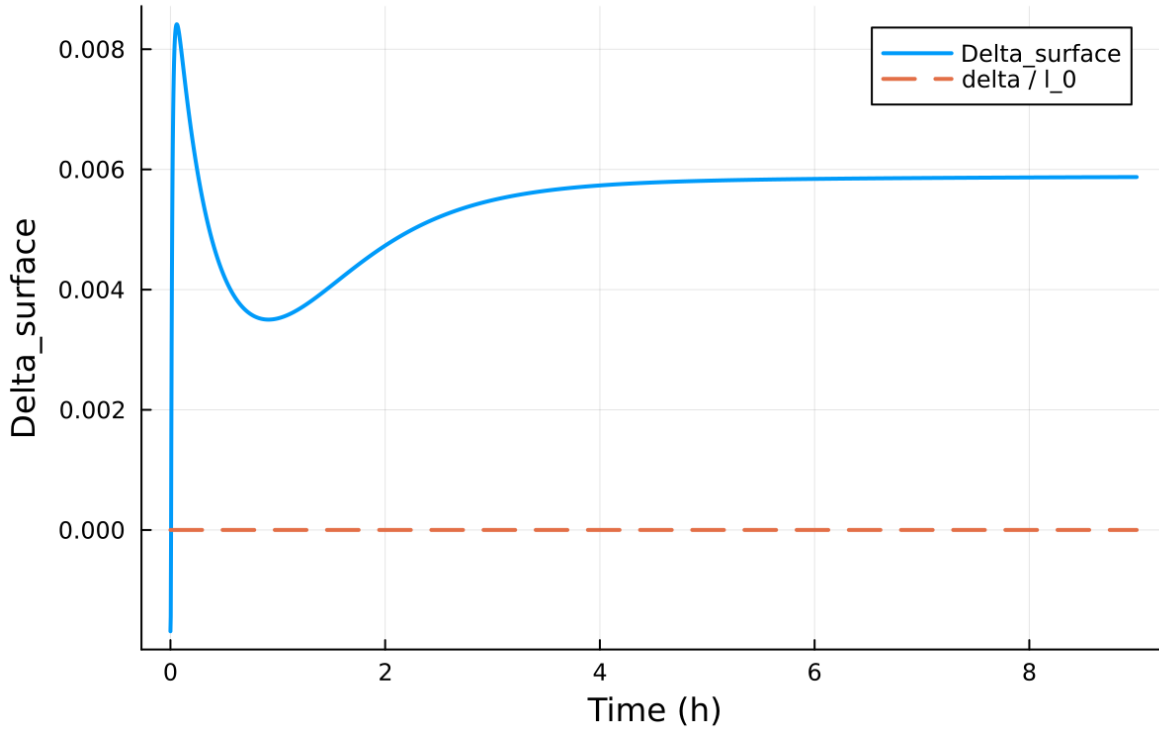


Figure 8: Fold proximity diagnostics versus time

6 Case 3: idealized sbl 120x1m 12h report

7 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/collapse_test/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver), Val{:forward}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

7.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/collapse_test`
- Number of saved time samples: 1441
- Number of profile snapshots: 25
- Solver accepted/rejected steps: 1073 / 7
- RHS evaluations: 398141
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

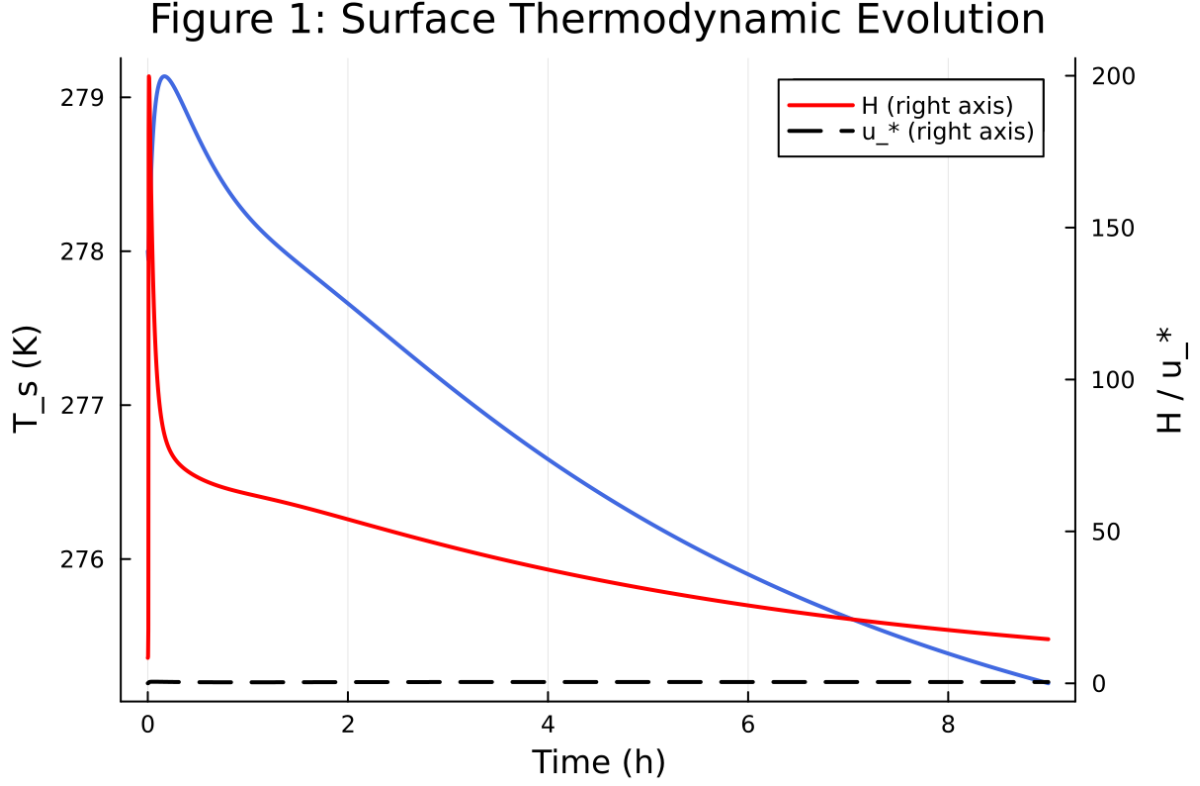


Figure 9: Time series of T_s , sensible heat flux, and friction velocity

7.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.048542, 106.702033] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.001272, 86.675008]$

7.3 Generated Artifacts

- Time-series CSV: `results/collapse_test/time_series.csv`
- Diagnostic summary JSON: `results/collapse_test/summary.json`
- Payload JLD2: `results/collapse_test/payload.jld2`

7.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

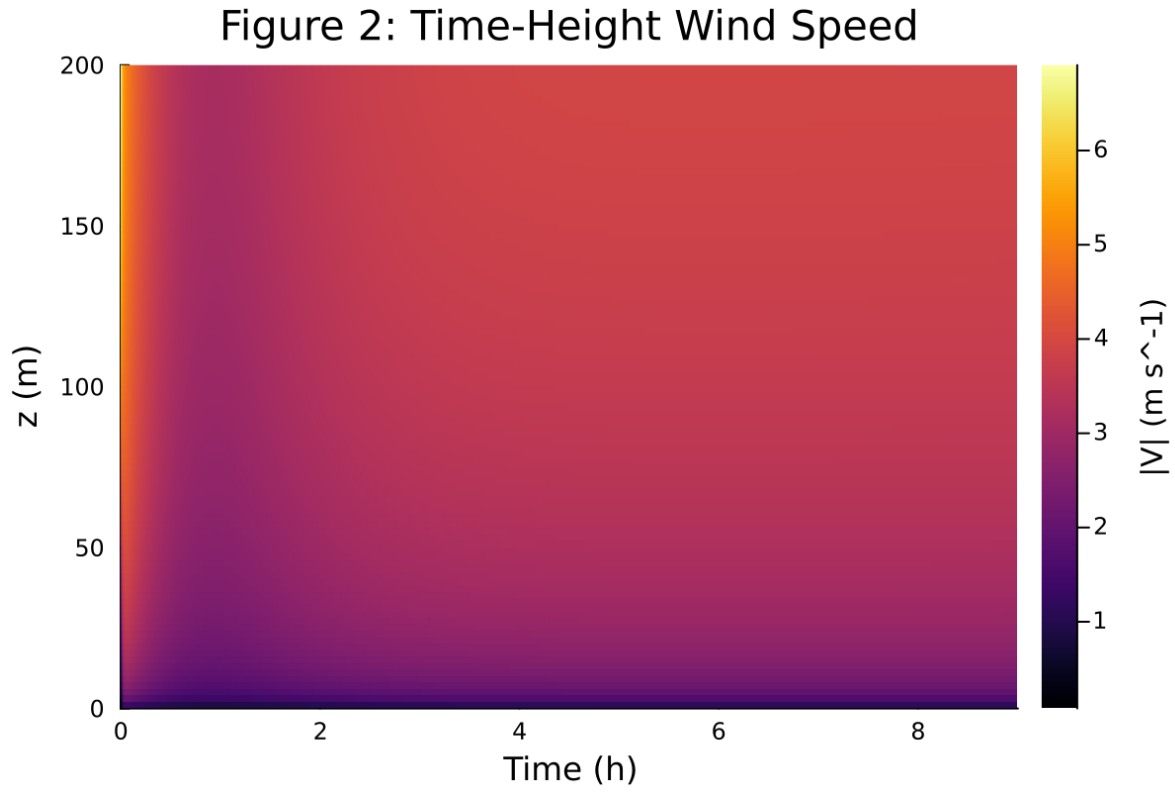


Figure 10: Time-height contour of wind speed with LLJ evolution

6. Phase portrait on regularized manifold (Delta vs e_{xi})
7. Eddy diffusivity response (Km, Kh) versus stability/Richardson number
8. Fold proximity diagnostics versus time

7.5 Included Plots

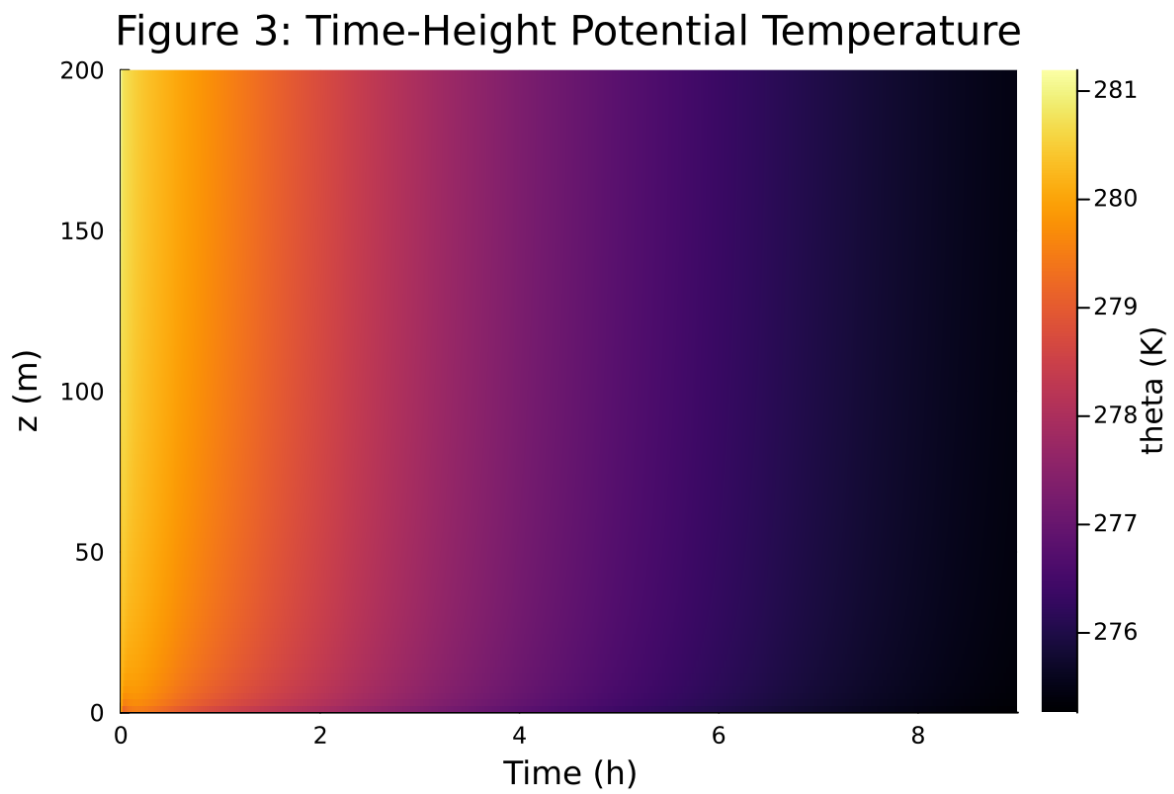


Figure 11: Time-height contour of potential temperature and inversion growth

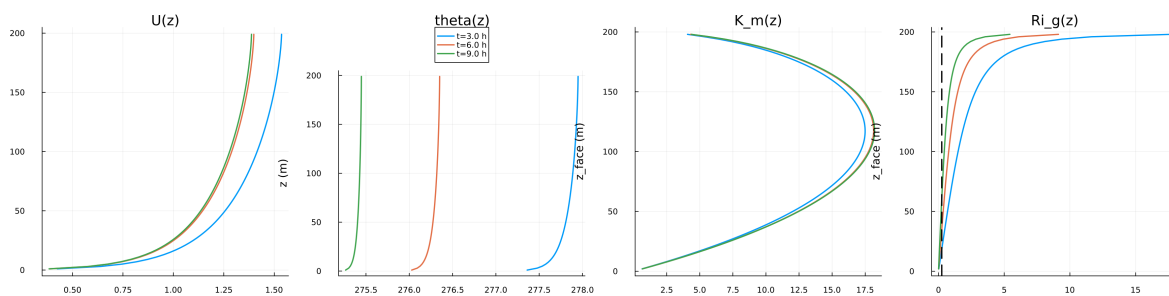


Figure 12: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

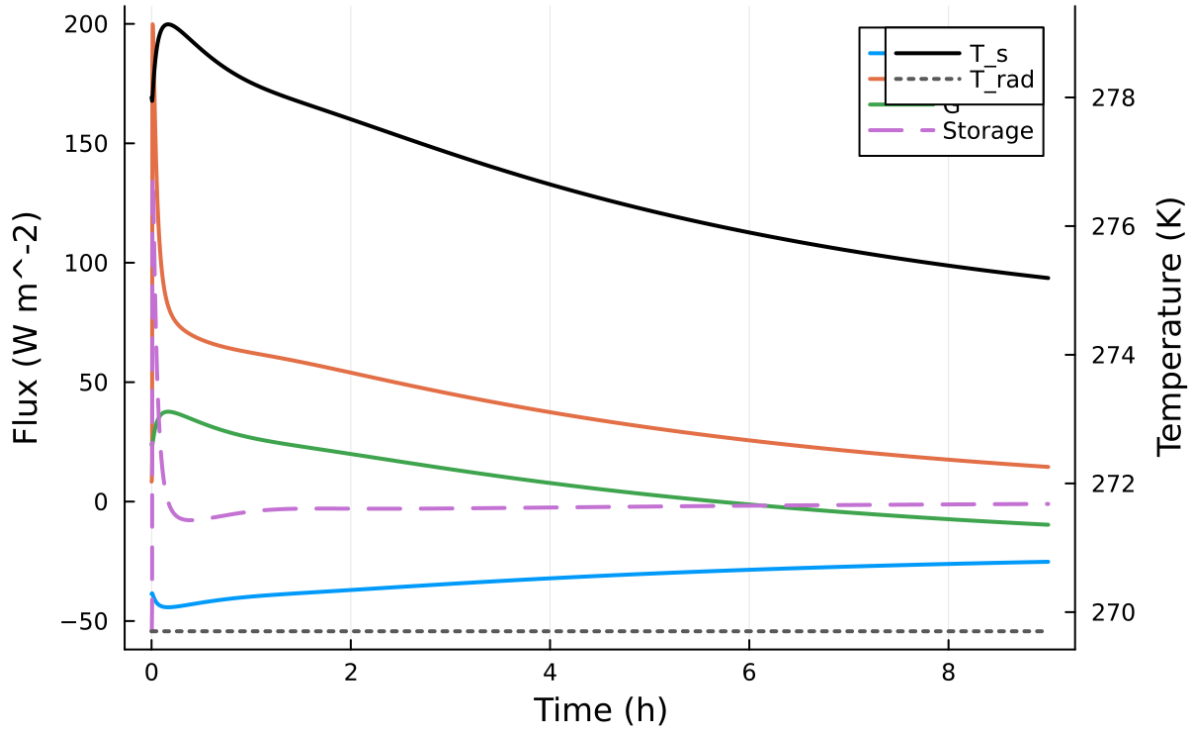


Figure 13: Surface energy budget components (R_n , H , G , storage)

8 Case 4: gabls1 80x2m 0h report

9 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`

. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}()}, true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.1` and returned `Success` .

9.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 62 / 1
- RHS evaluations: 15572
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 6: Regularized Slow-Manifold Phase Portrait

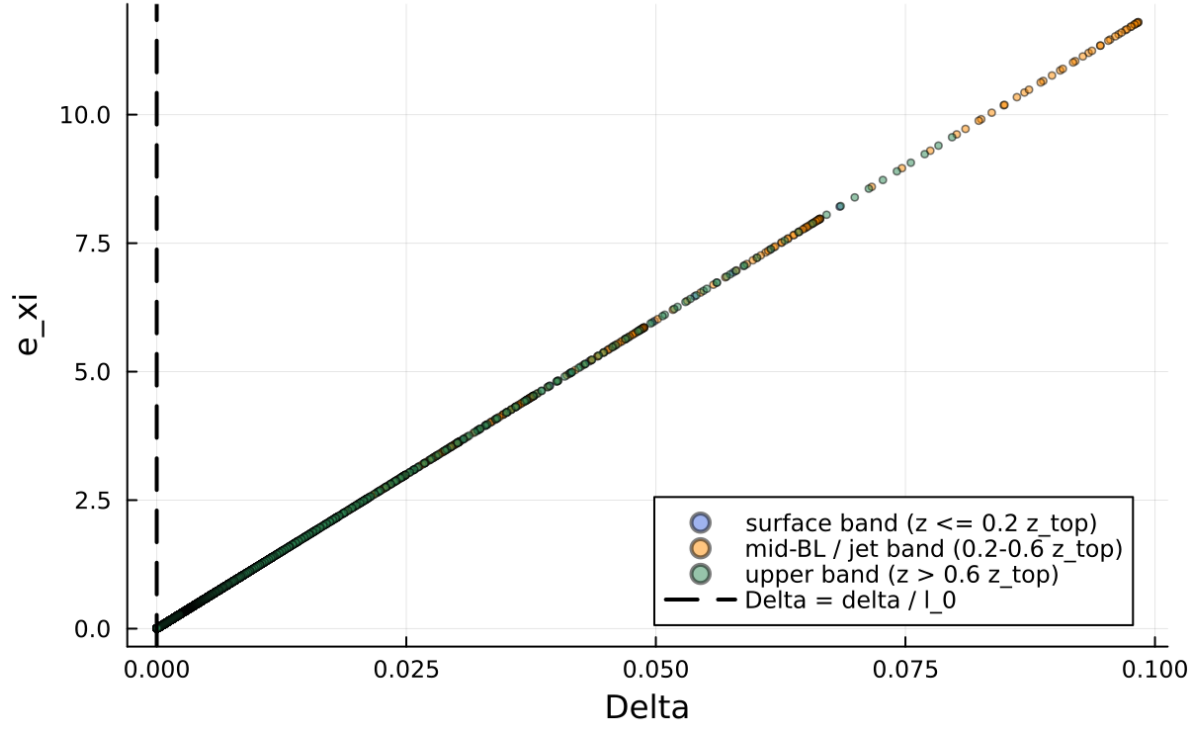


Figure 14: Phase portrait on regularized manifold (Δ vs e_{xi})

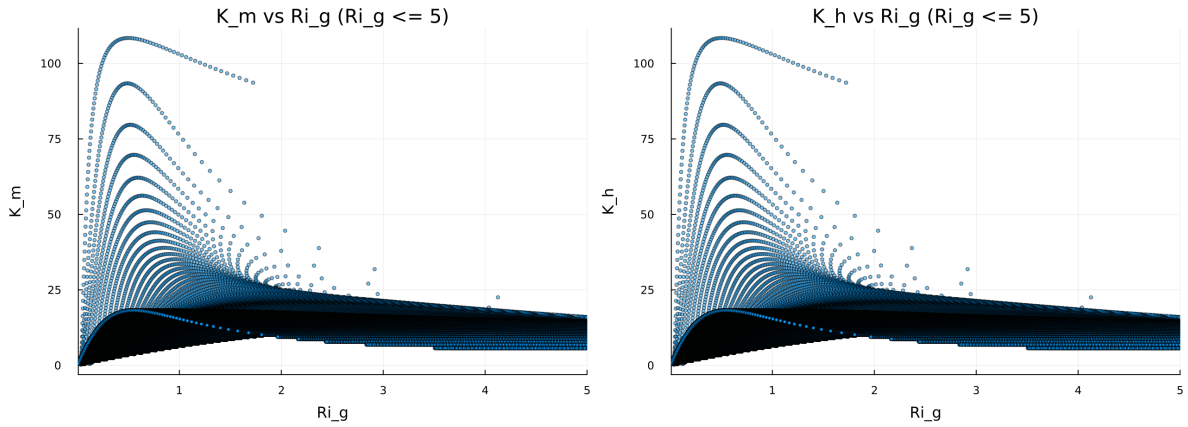


Figure 15: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

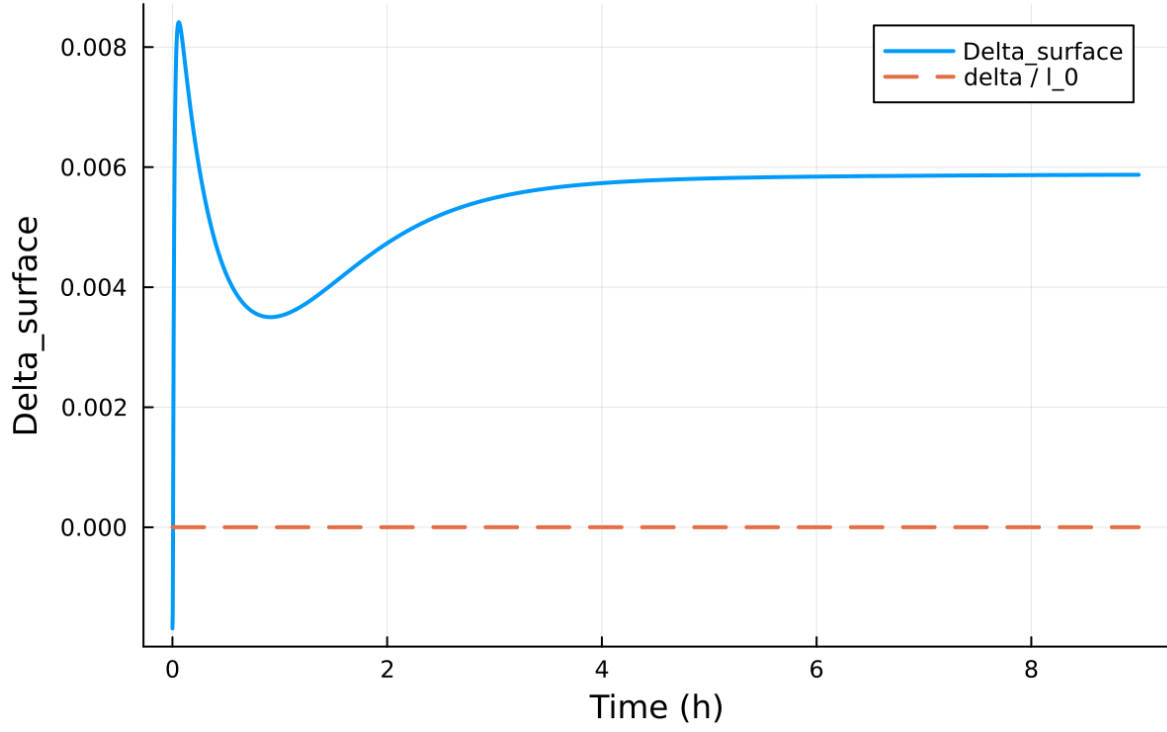


Figure 16: Fold proximity diagnostics versus time

Table 1: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

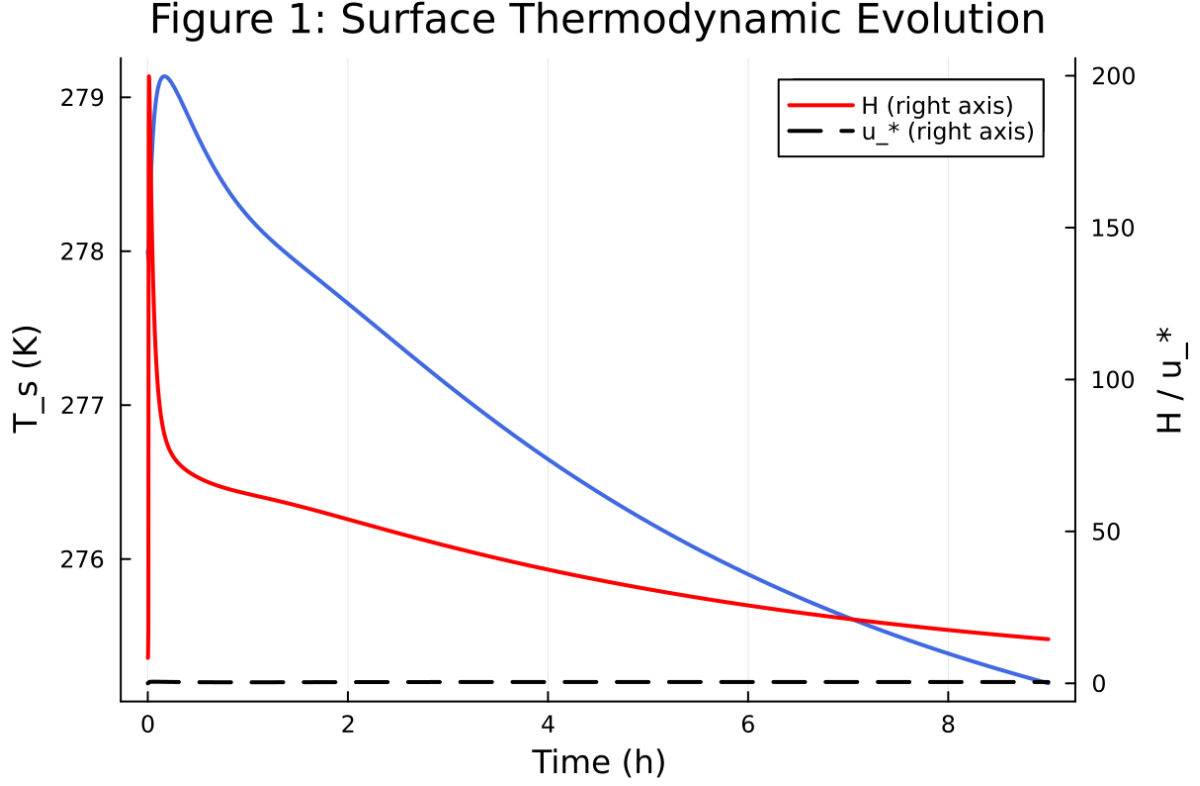


Figure 17: Time series of T_s , sensible heat flux, and friction velocity

9.2 Forcing and Boundary Conditions

9.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.119669, 36.559392]$

9.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

9.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

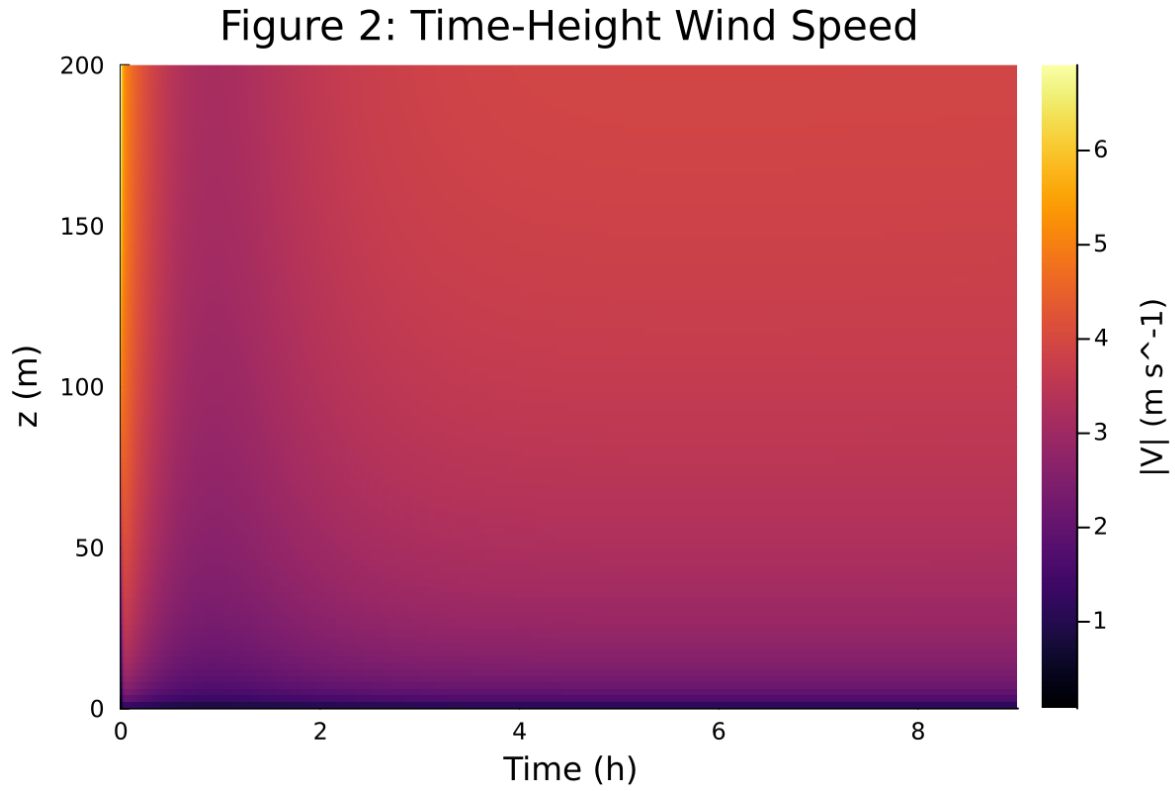


Figure 18: Time-height contour of wind speed with LLJ evolution

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

9.6 Included Plots

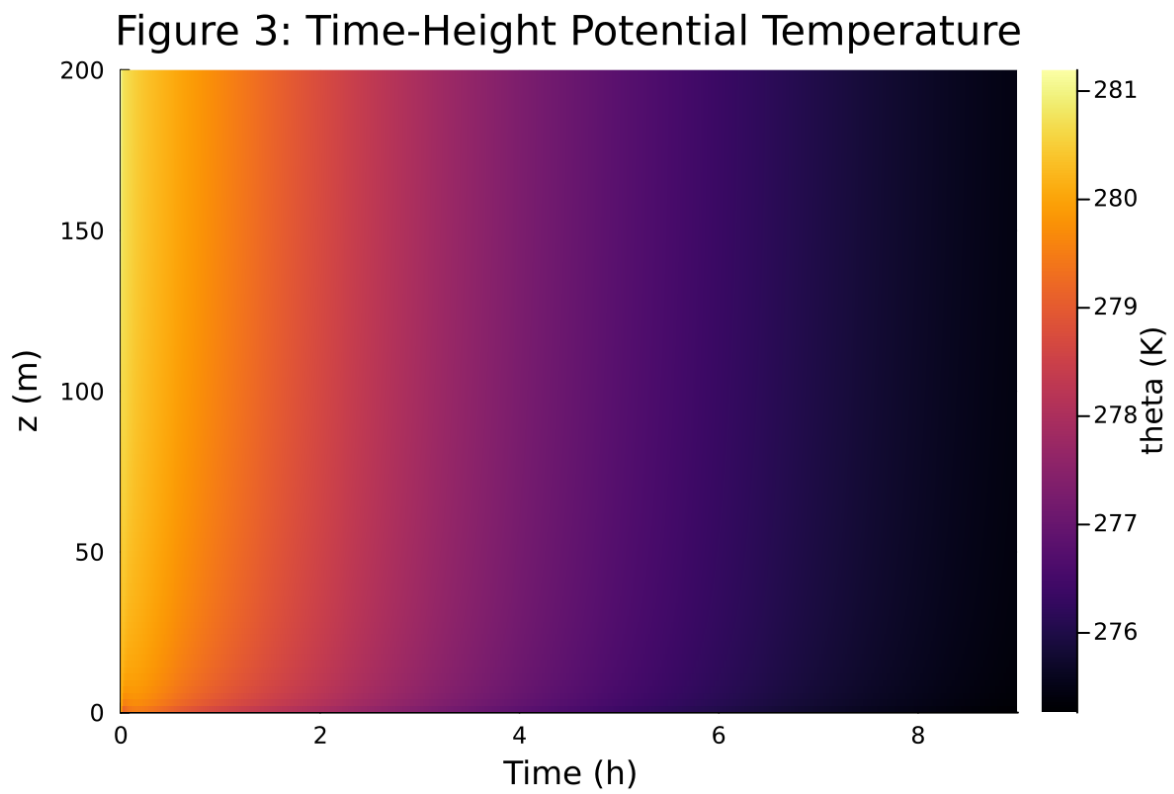


Figure 19: Time-height contour of potential temperature and inversion growth

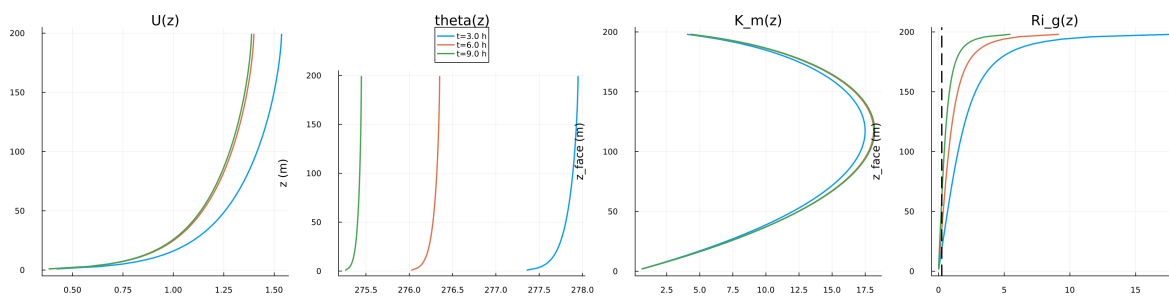


Figure 20: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

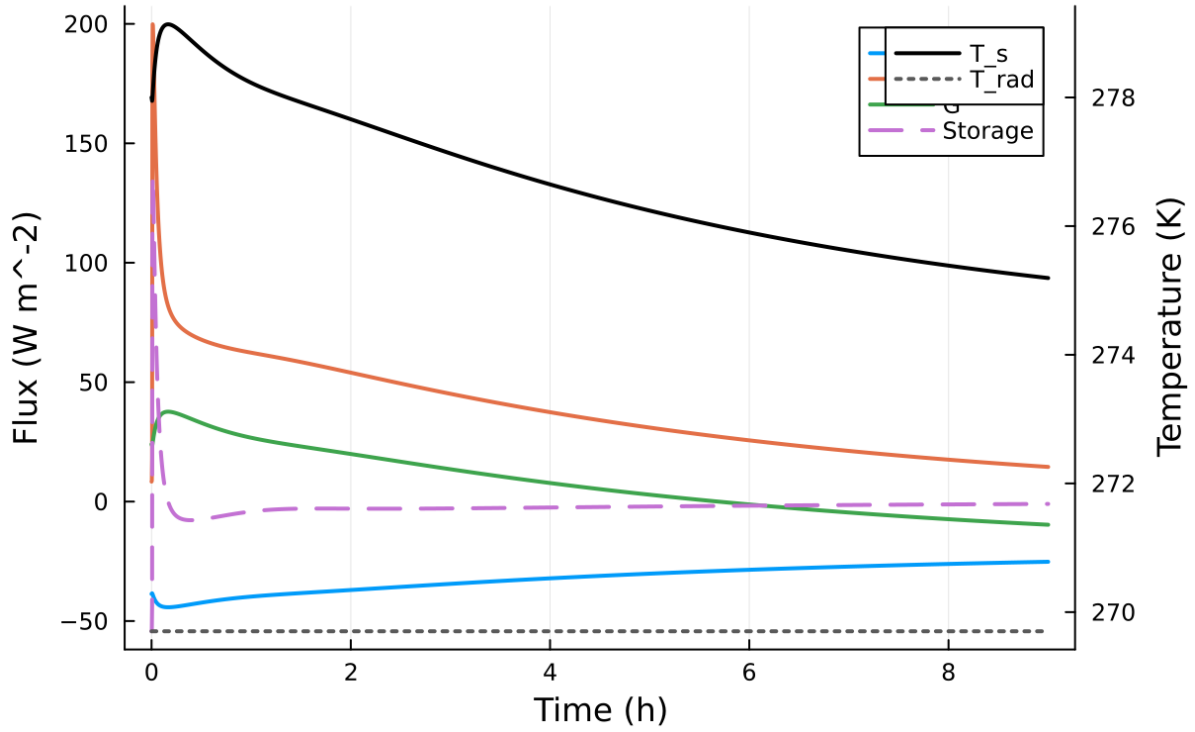


Figure 21: Surface energy budget components (R_n , H , G , storage)

10 Case 5: gabls1 80x2m 9h report

11 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`

. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.1` and returned `Success` .

11.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 416 / 1
- RHS evaluations: 104426
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 6: Regularized Slow-Manifold Phase Portrait

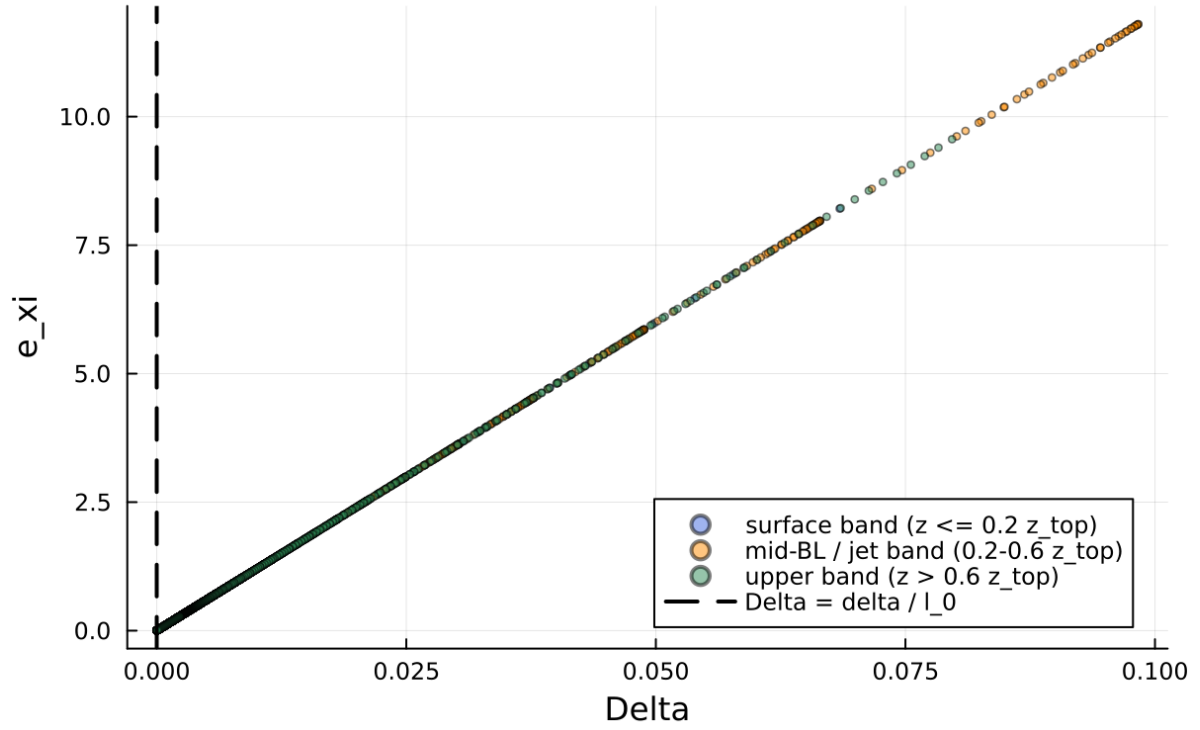


Figure 22: Phase portrait on regularized manifold (Δ vs e_{xi})

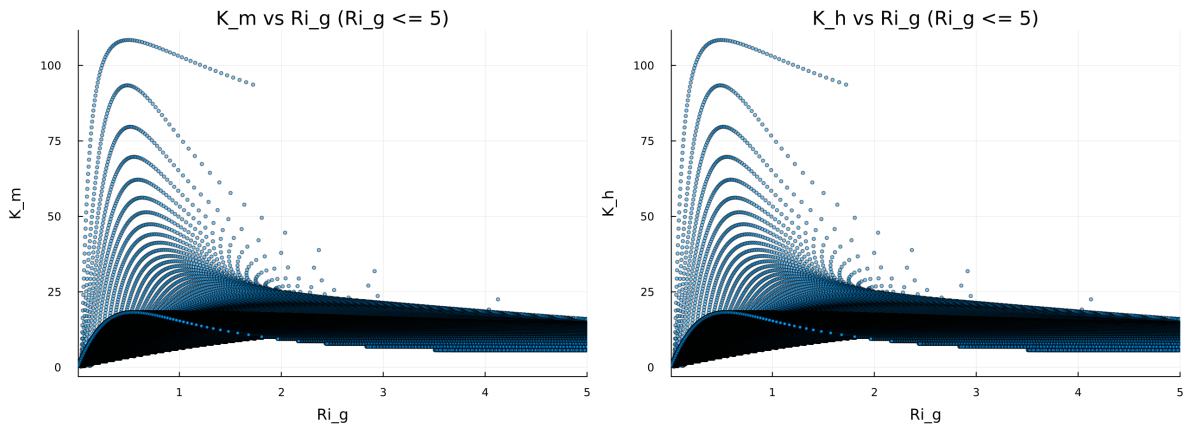


Figure 23: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

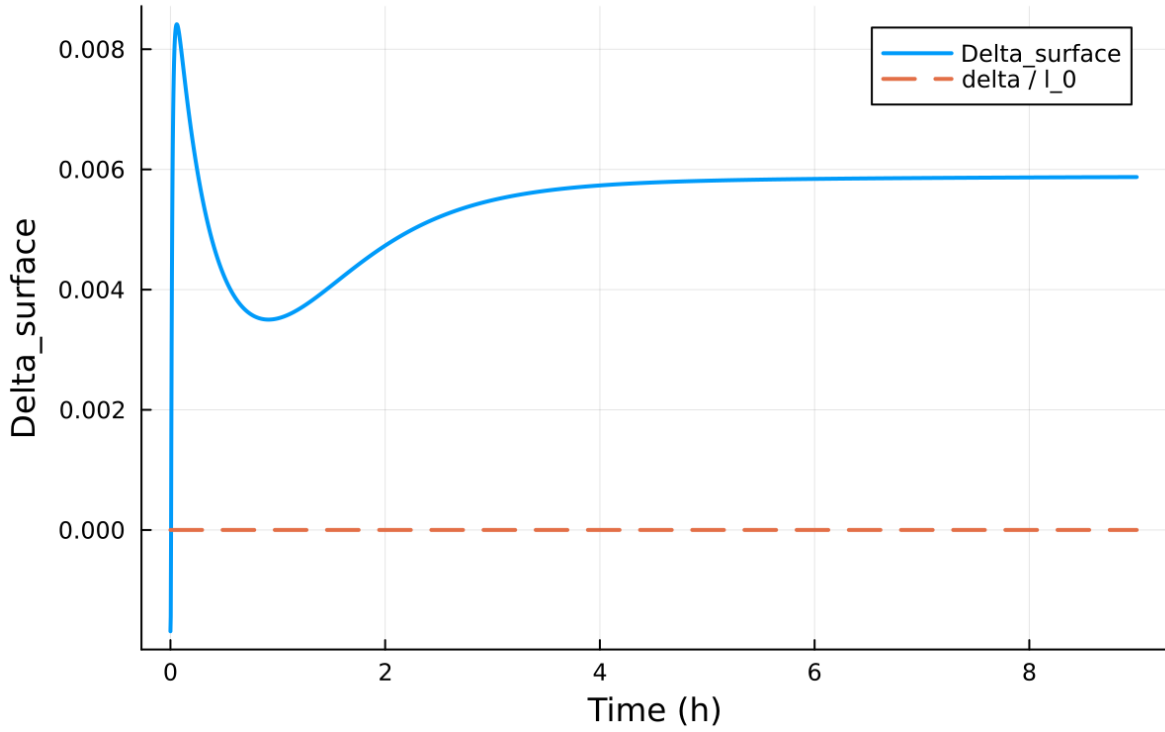


Figure 24: Fold proximity diagnostics versus time

11.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.009840, 82.319866]$

11.3 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

11.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

Figure 1: Surface Thermodynamic Evolution

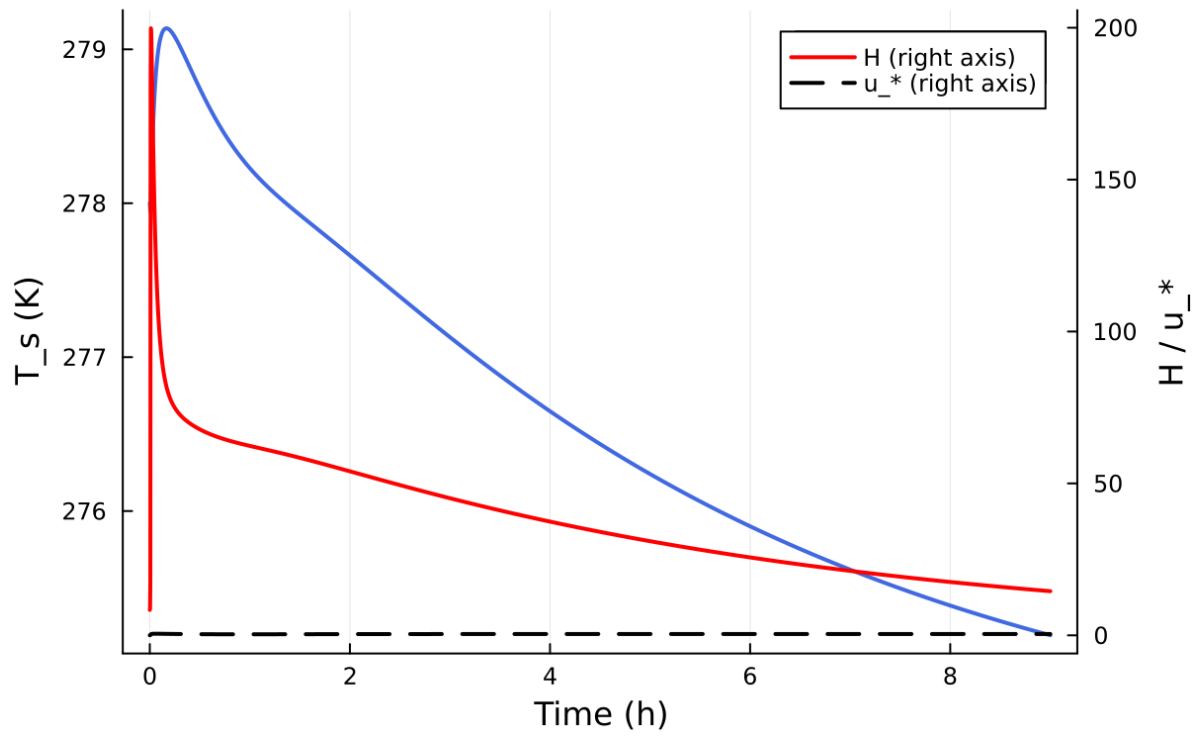


Figure 25: Time series of T_s , sensible heat flux, and friction velocity

6. Phase portrait on regularized manifold (Delta vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

11.5 Included Plots

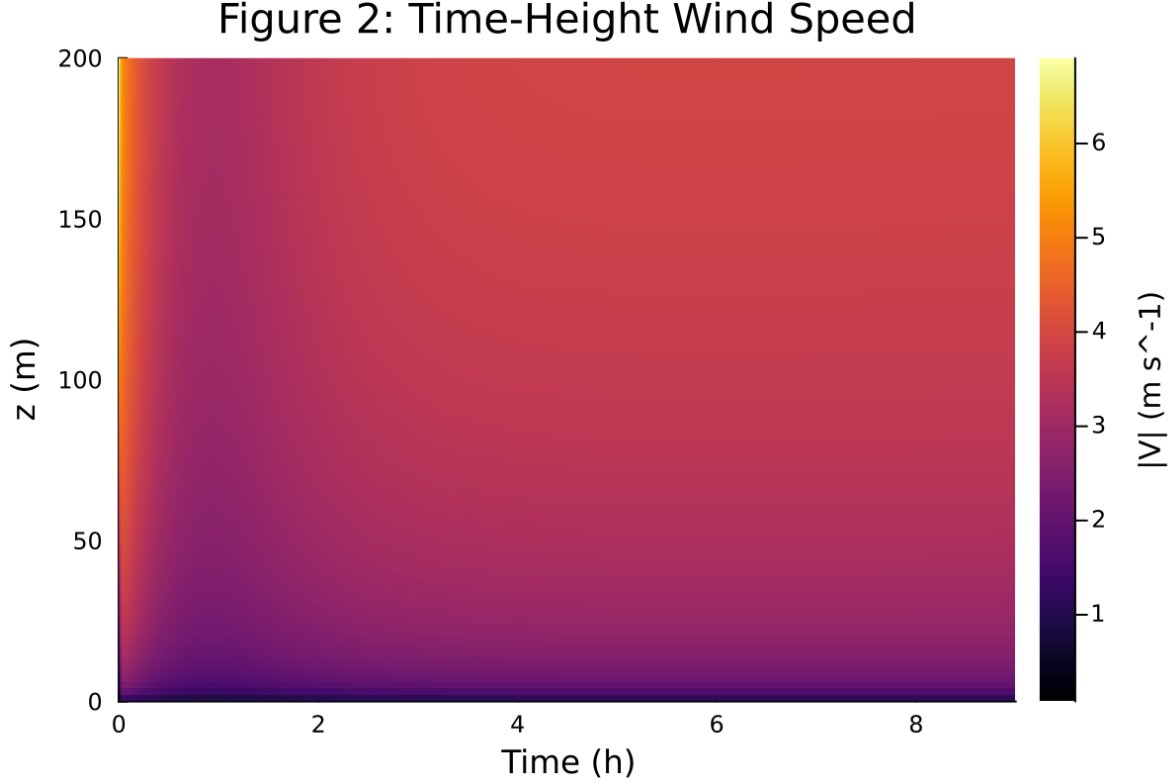


Figure 26: Time-height contour of wind speed with LLJ evolution

12 Case 6: scm case report

13 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/gabls1/summary.json`

. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t{` and returned `Success` .

13.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/gabls1`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 62 / 1
- RHS evaluations: 15572
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

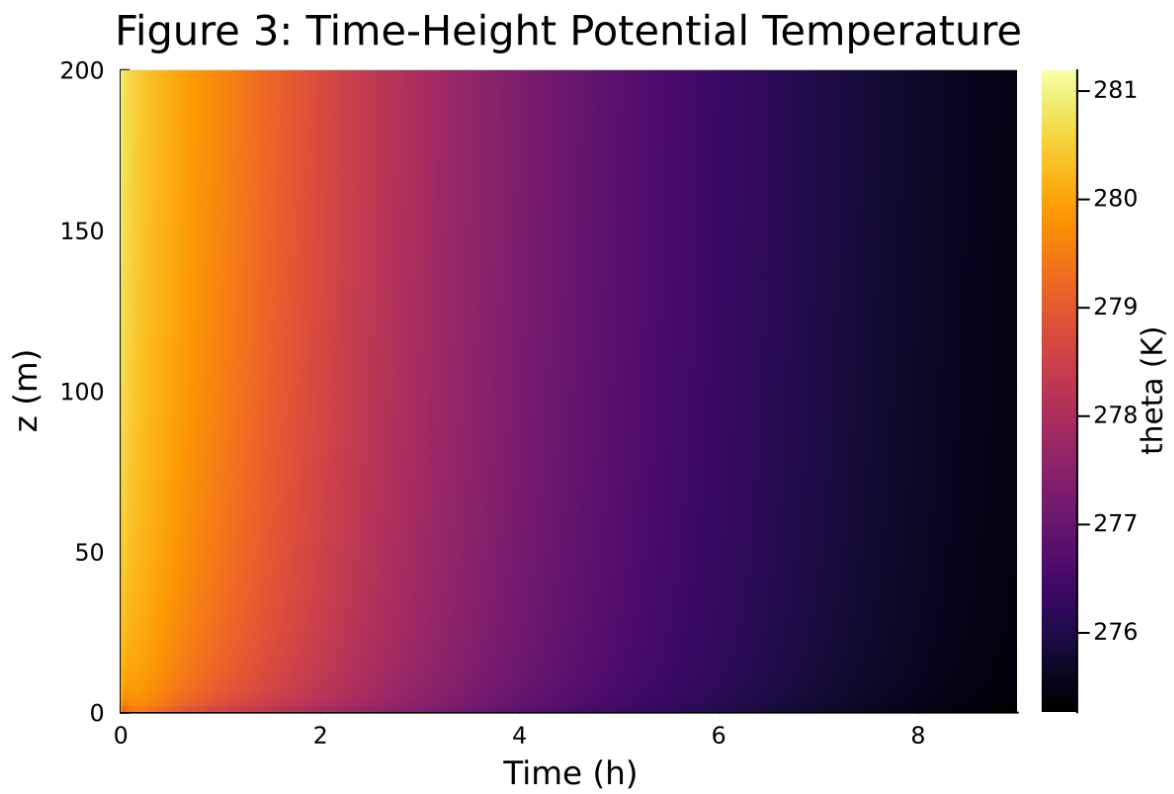


Figure 27: Time-height contour of potential temperature and inversion growth

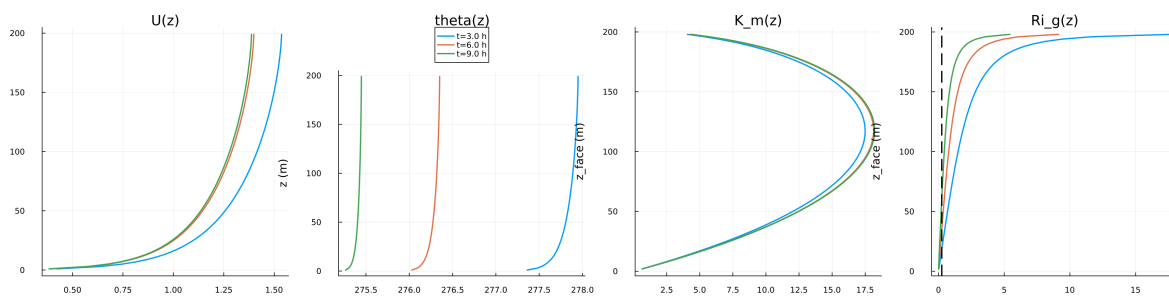


Figure 28: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

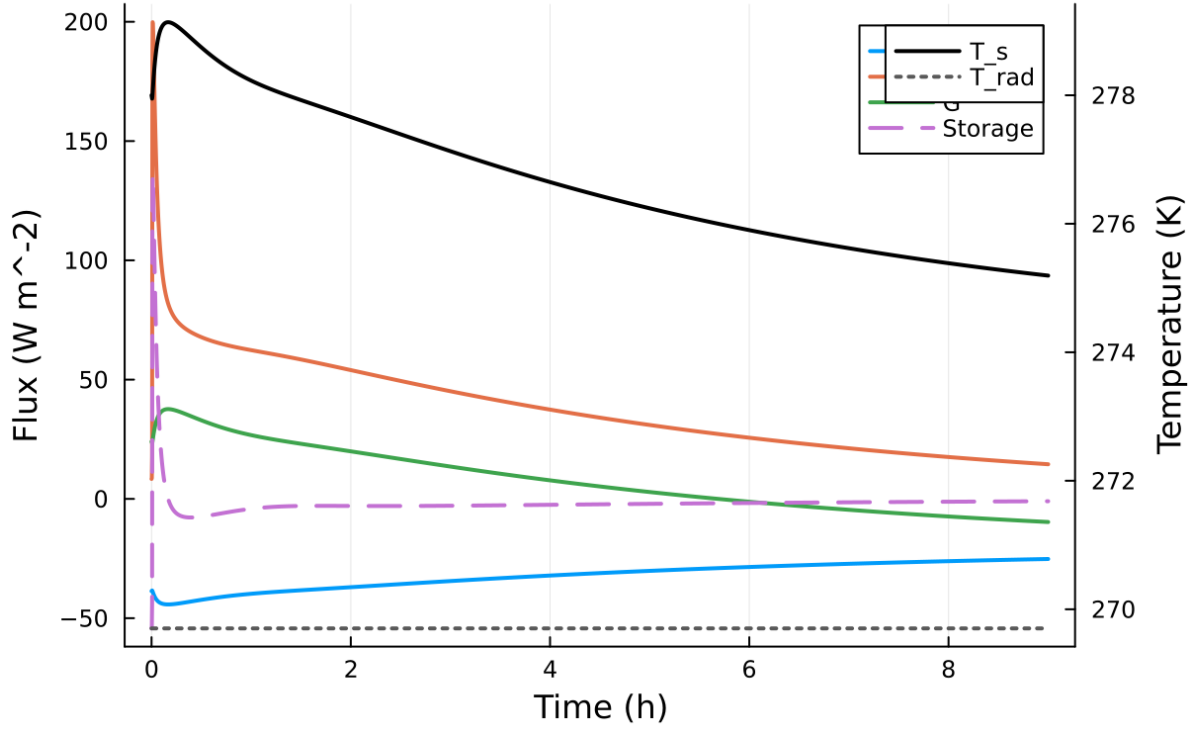


Figure 29: Surface energy budget components (R_n , H , G , storage)

Table 2: Forcing and Boundary Condition Parameters

Parameter	Symbol	Value
Zonal geostrophic wind	U_g	$8.000000 \text{ m s}^{-1}$
Meridional geostrophic wind	V_g	$0.000000 \text{ m s}^{-1}$
Surface roughness (momentum)	z_{0m}	n/a m
Surface roughness (heat)	z_{0h}	n/a m
Coriolis parameter	f	$1.000\text{e-}04 \text{ s}^{-1}$
Reference potential temperature	θ_a	265.000000 K
Deep soil temperature	T_{deep}	262.000000 K
Downward longwave forcing	$R_{\downarrow}$	$245.000000 \text{ W m}^{-2}$
Soil conductivity proxy	λ_s	1.200000
Soil slab depth	d_{soil}	0.100000 m
Surface diffusivity floor	$k_{min,surf}$	$0.001000 \text{ m}^2 \text{ s}^{-1}$
Top thermal BC type	—	neumann
Top boundary potential temperature	θ_{top}	265.000000 K
Top relaxation coefficient	λ_{top}	0.000000 s^{-1}

Figure 6: Regularized Slow-Manifold Phase Portrait

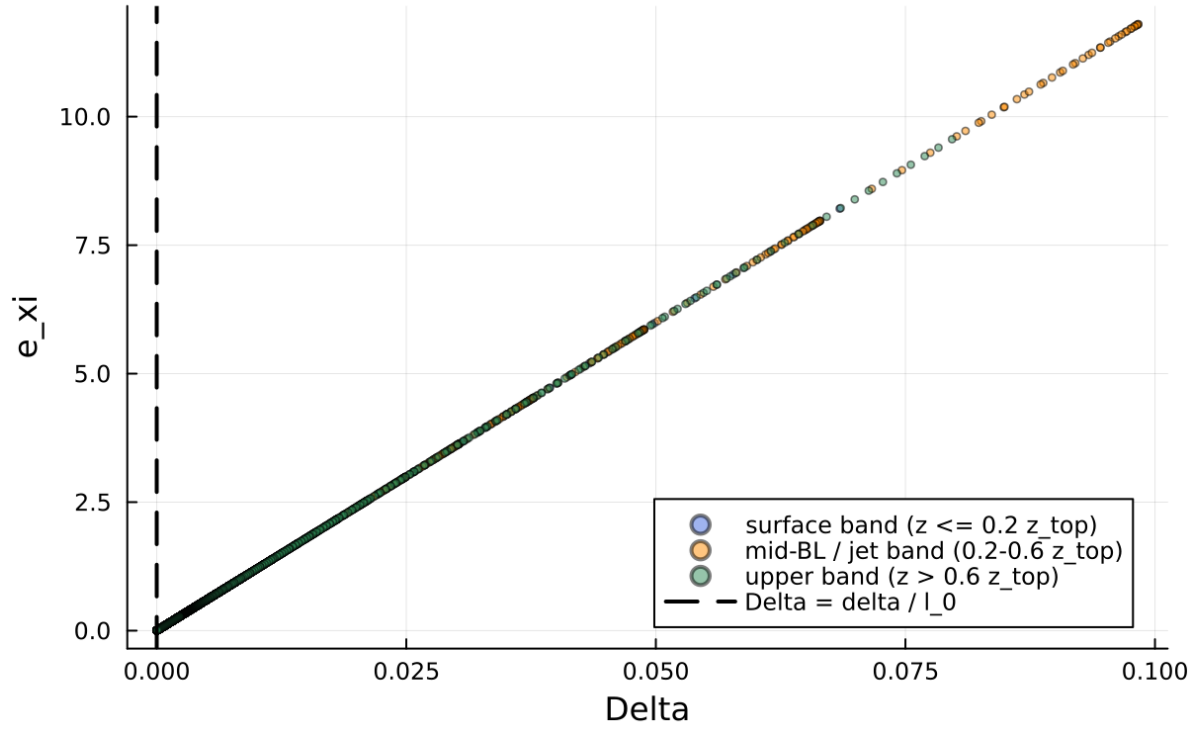


Figure 30: Phase portrait on regularized manifold (Δ vs e_{xi})

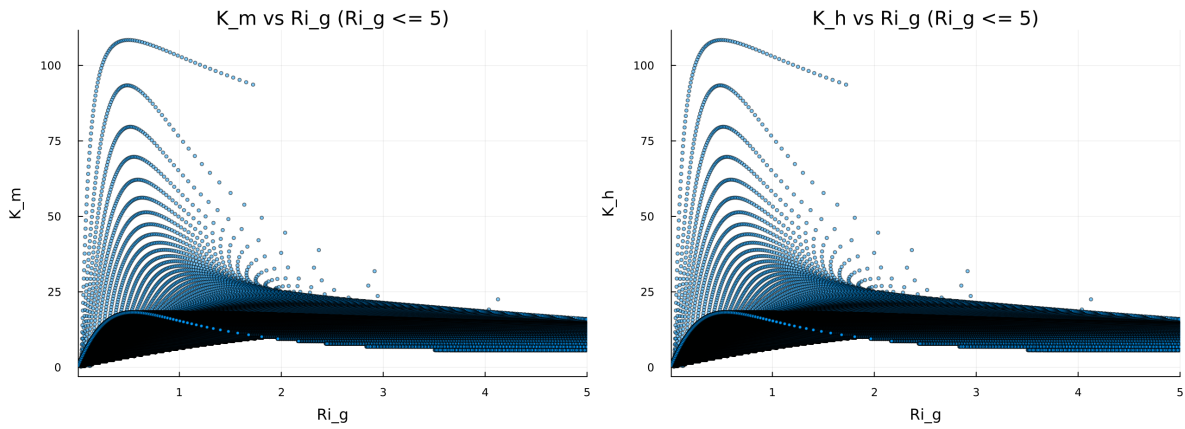


Figure 31: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

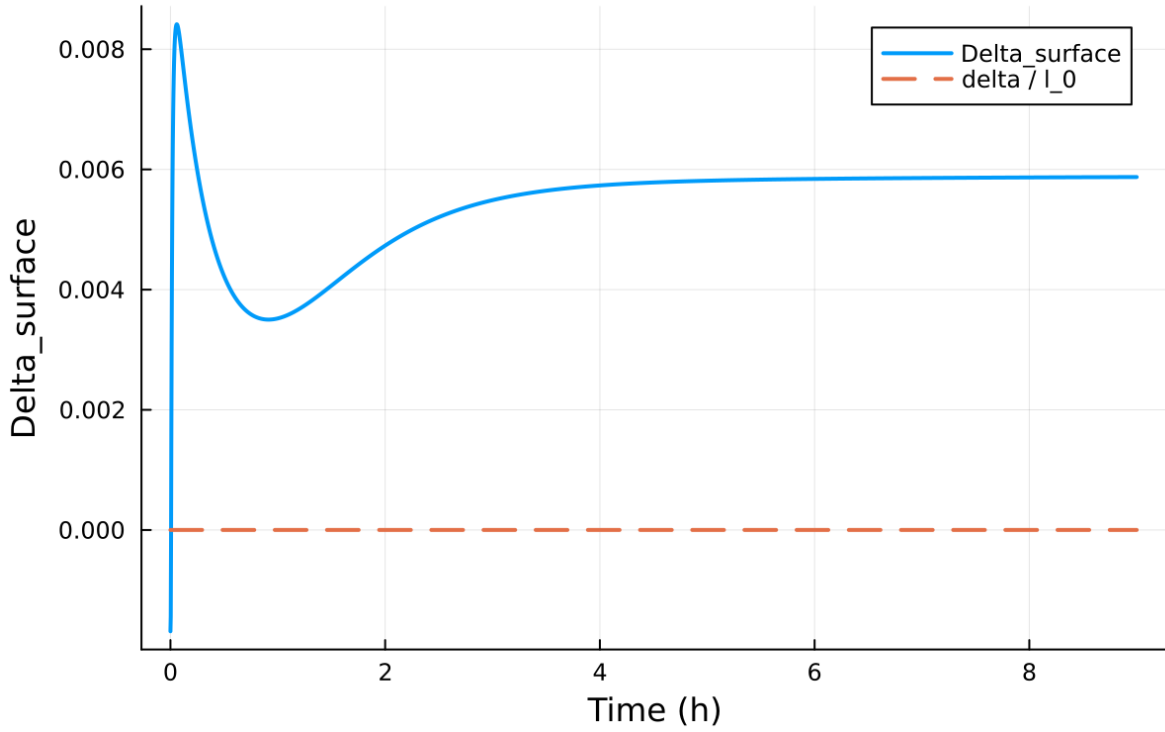


Figure 32: Fold proximity diagnostics versus time

13.2 Forcing and Boundary Conditions

13.3 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.119669, 36.559392]$

13.4 Generated Artifacts

- Time-series CSV: `results/gabls1/time_series.csv`
- Diagnostic summary JSON: `results/gabls1/summary.json`
- Payload JLD2: `results/gabls1/payload.jld2`

13.5 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times

Figure 1: Surface Thermodynamic Evolution

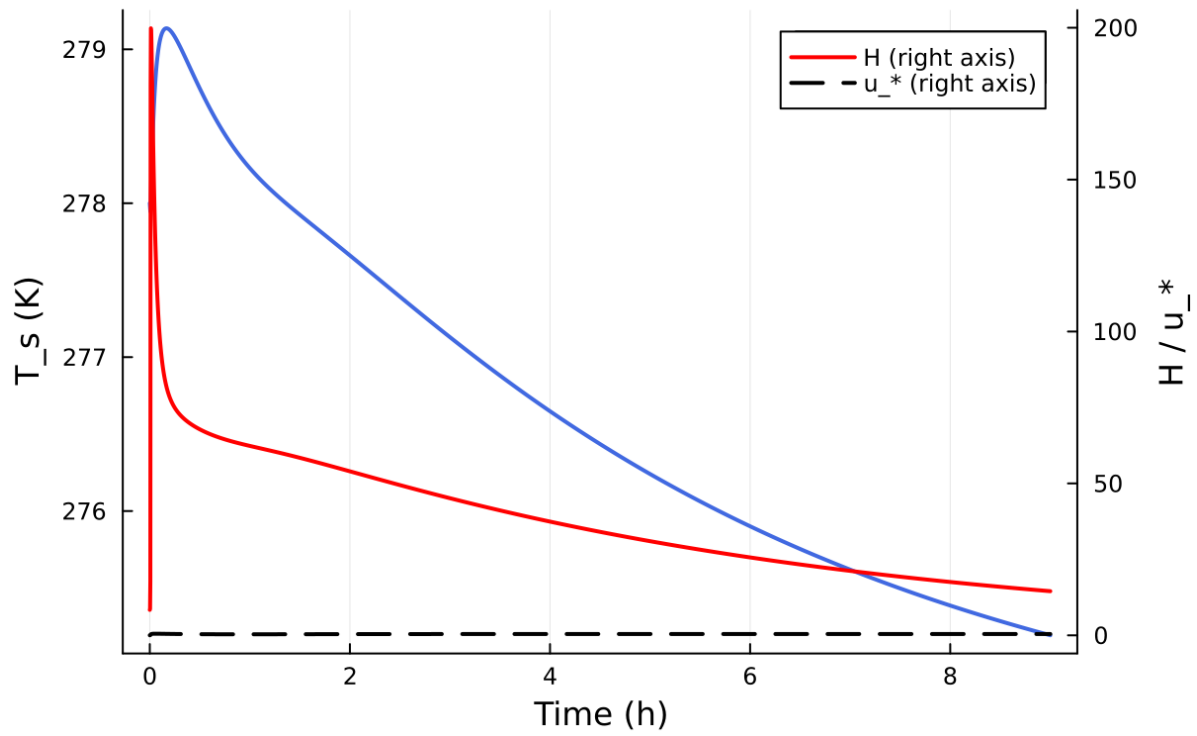


Figure 33: Time series of T_s , sensible heat flux, and friction velocity

5. Surface energy budget components (R_n , H , G , storage)
6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

13.6 Included Plots

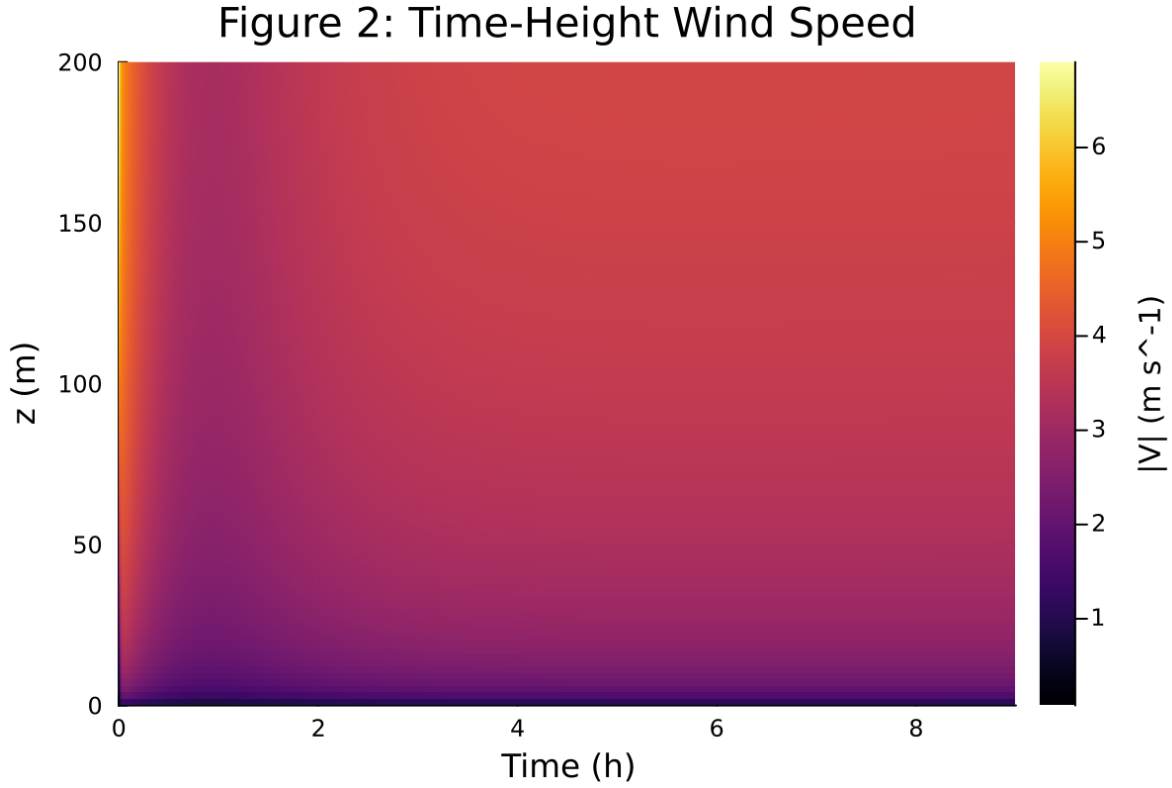


Figure 34: Time-height contour of wind speed with LLJ evolution

14 Case 7: idealized sbl 80x2m 9h report

15 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PRE{Val{:forward}}(), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.t` and returned `Success`.

15.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl`
- Number of saved time samples: 1081
- Number of profile snapshots: 19
- Solver accepted/rejected steps: 438 / 2
- RHS evaluations: 109956
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

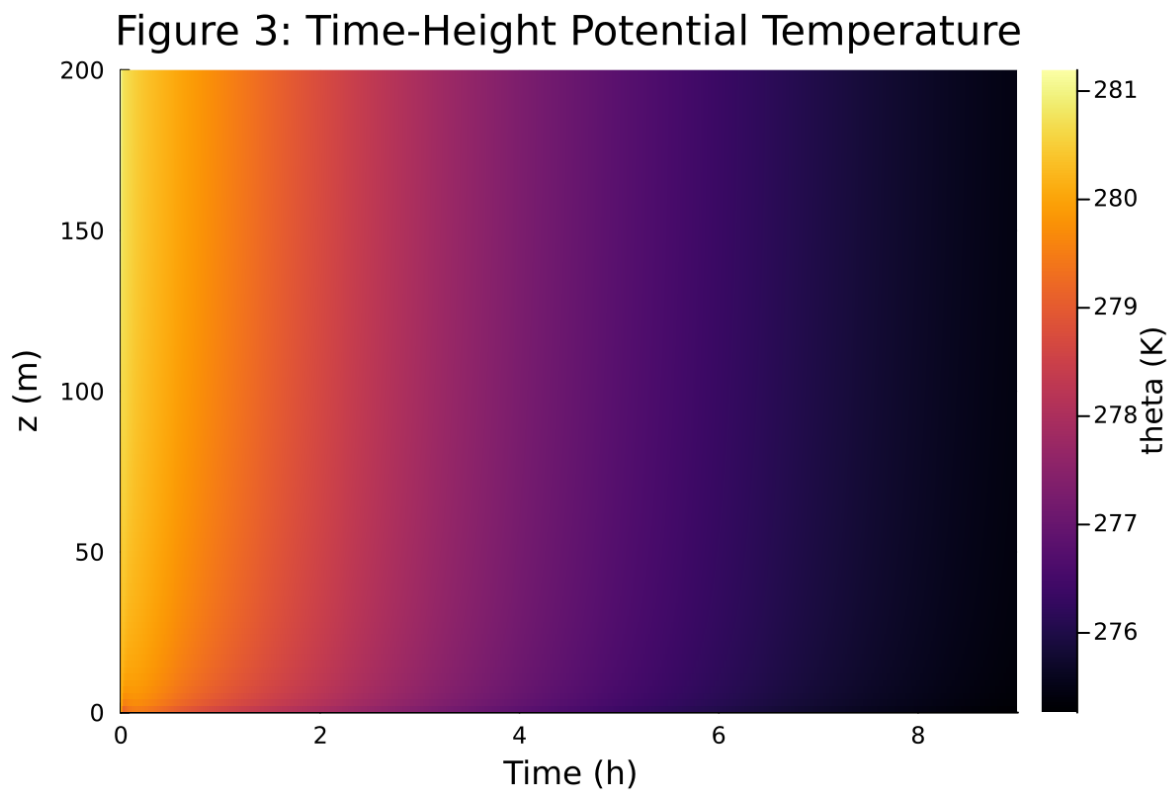


Figure 35: Time-height contour of potential temperature and inversion growth

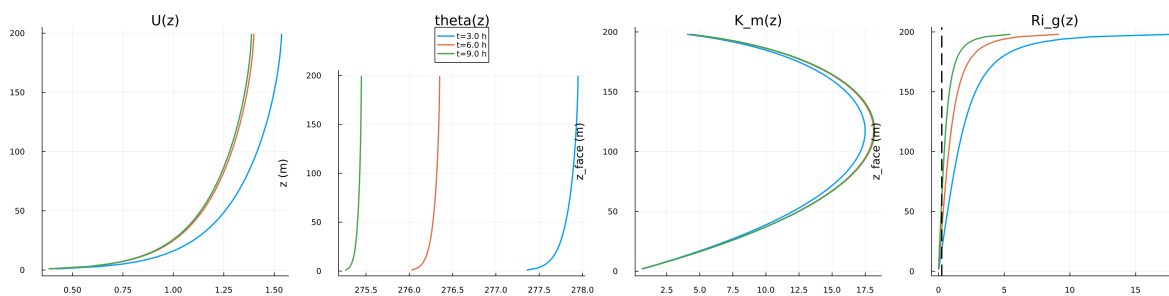


Figure 36: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

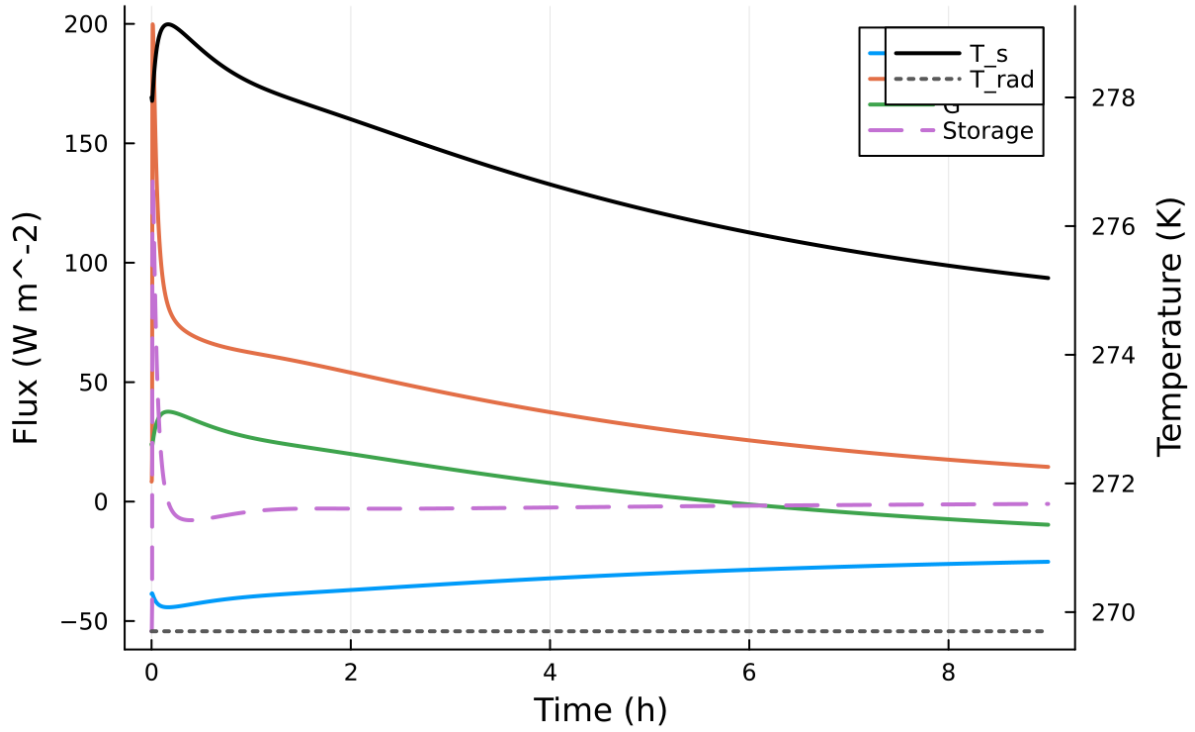


Figure 37: Surface energy budget components (R_n , H , G , storage)

15.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.007283, 43.999865]$

15.3 Generated Artifacts

- Time-series CSV: `results/idealized_sbl/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl/summary.json`
- Payload JLD2: `results/idealized_sbl/payload.jld2`

15.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

Figure 6: Regularized Slow-Manifold Phase Portrait

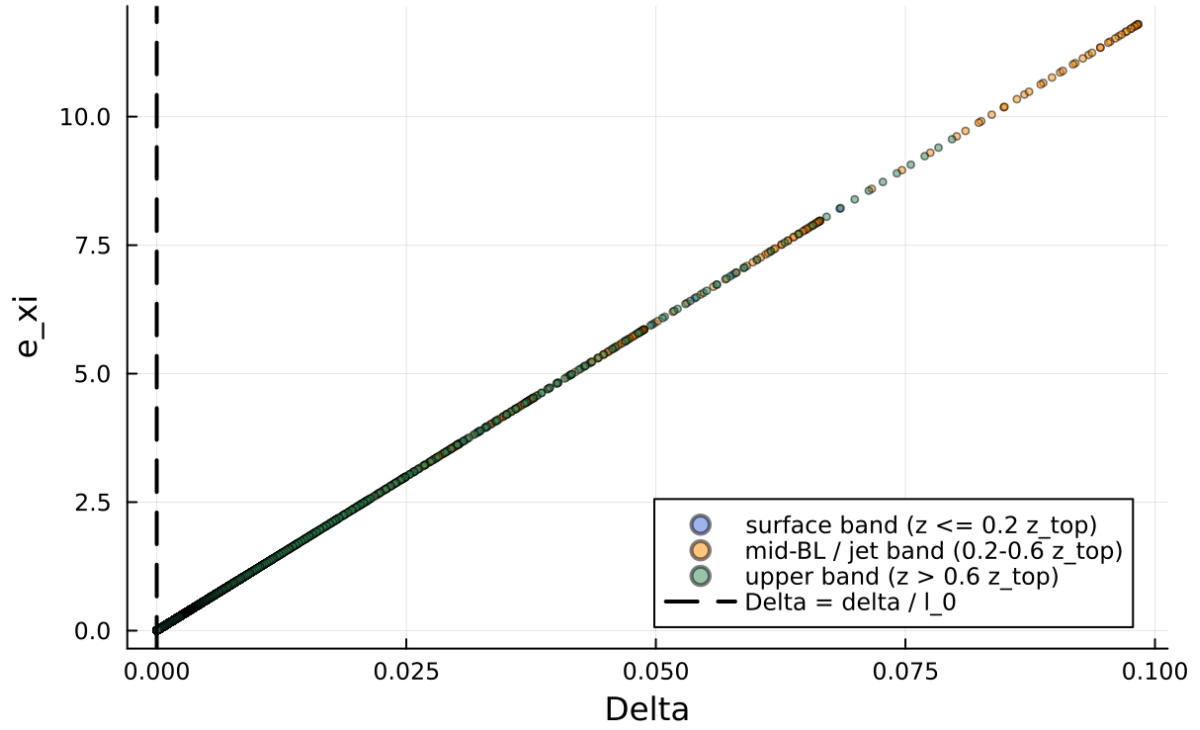


Figure 38: Phase portrait on regularized manifold (Δ vs e_{xi})

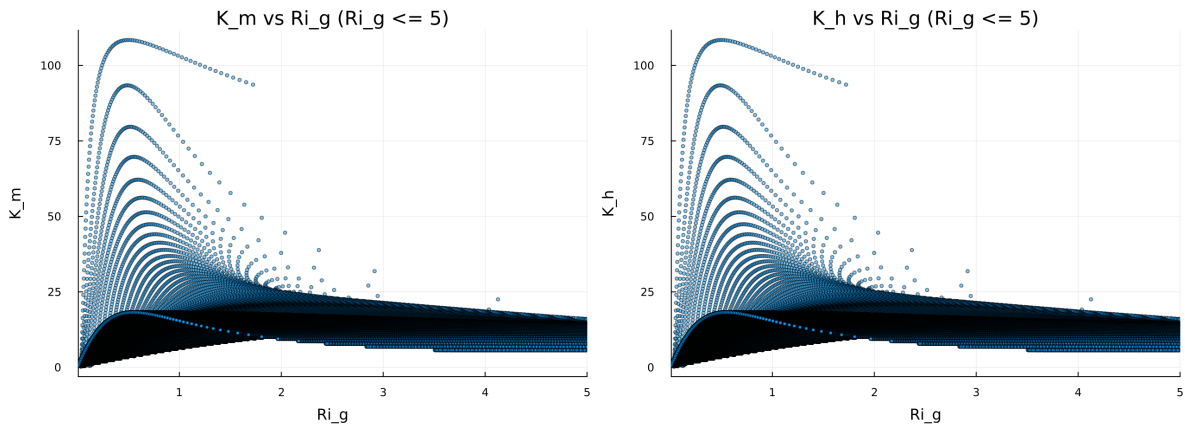


Figure 39: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

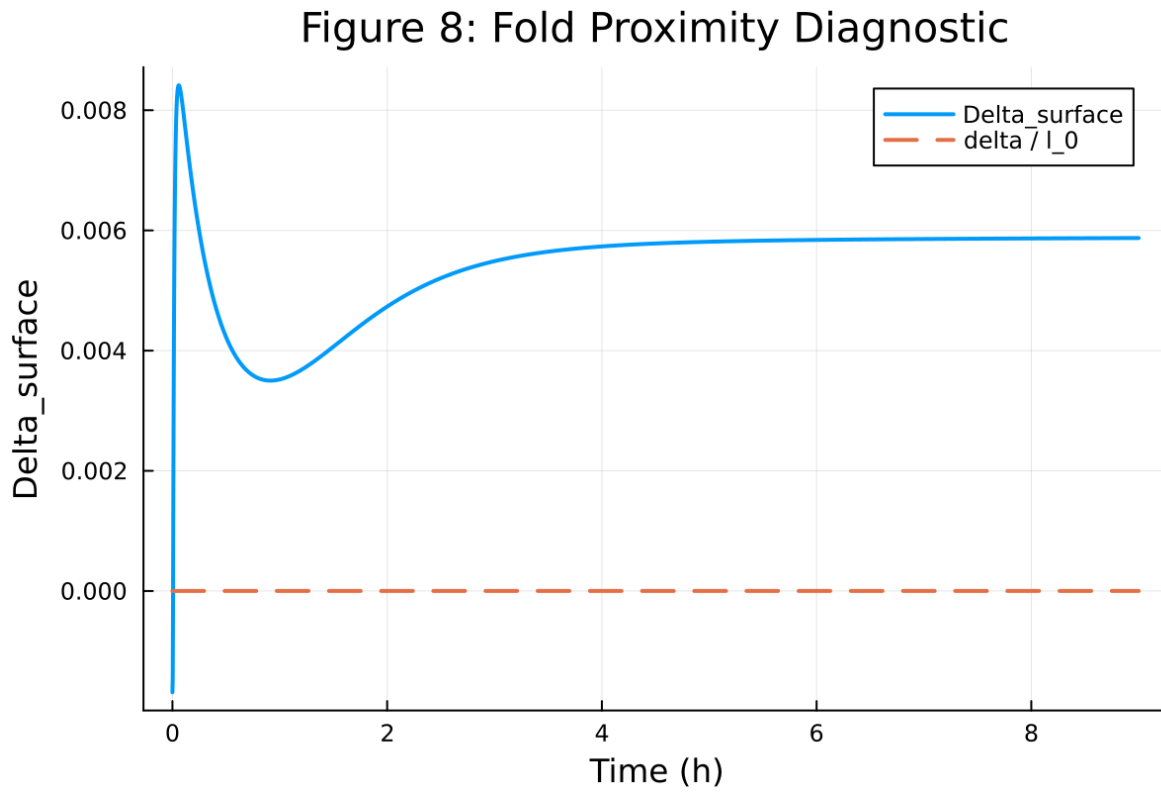


Figure 40: Fold proximity diagnostics versus time

6. Phase portrait on regularized manifold (Delta vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

15.5 Included Plots

16 Case 8: idealized sbl 80x2m 0h report

This section was generated automatically from the SCM diagnostics artifact `results/idealized_sbl_smoke/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver{Val{:forward}}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

- Case identifier: `idealized_sbl`
- Output directory: `results/idealized_sbl_smoke`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 65 / 2
- RHS evaluations: 16333
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reltol} = 1.000 \times 10^{-6}$

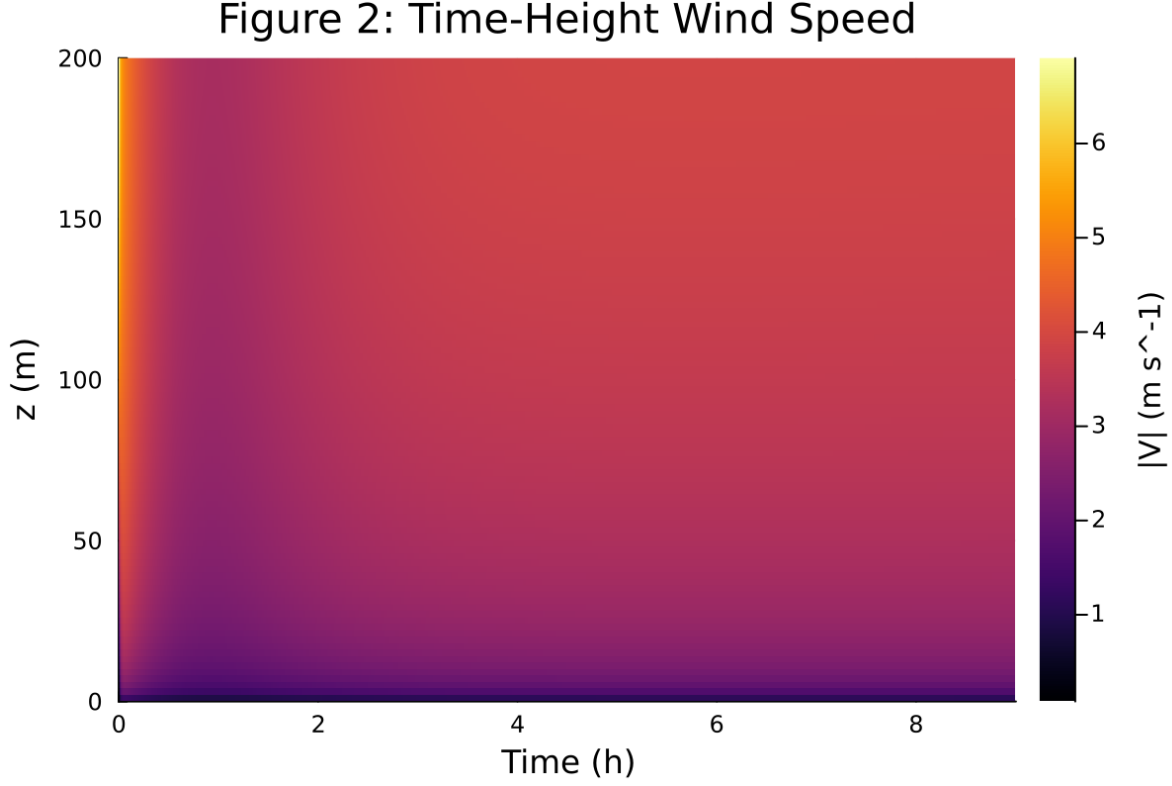


Figure 42: Time-height contour of wind speed with LLJ evolution

17.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.184562, 108.426452] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.034185, 34.722193]$

17.3 Generated Artifacts

- Time-series CSV: `results/idealized_sbl_smoke/time_series.csv`
- Diagnostic summary JSON: `results/idealized_sbl_smoke/summary.json`
- Payload JLD2: `not generated`

17.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

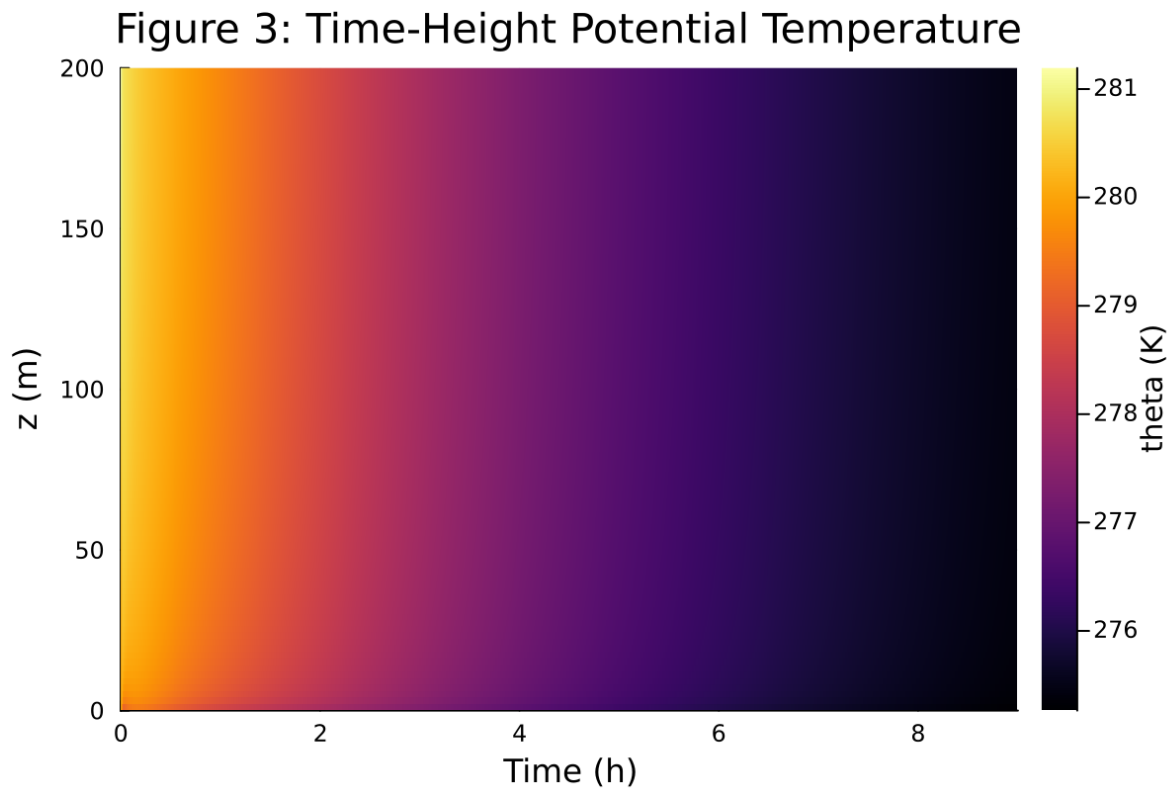


Figure 43: Time-height contour of potential temperature and inversion growth

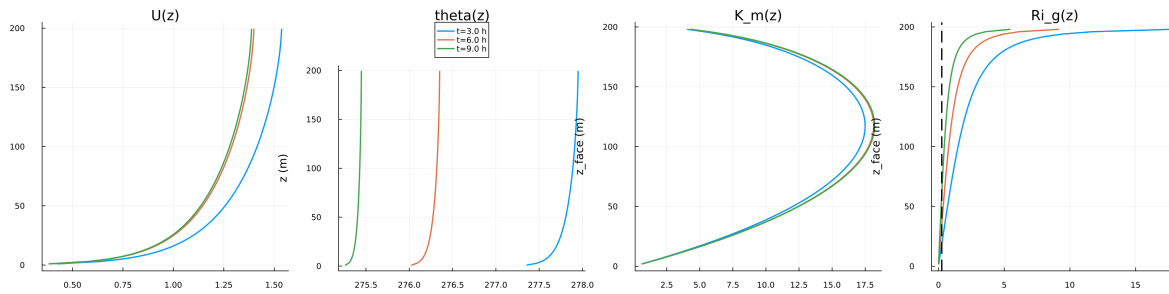


Figure 44: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

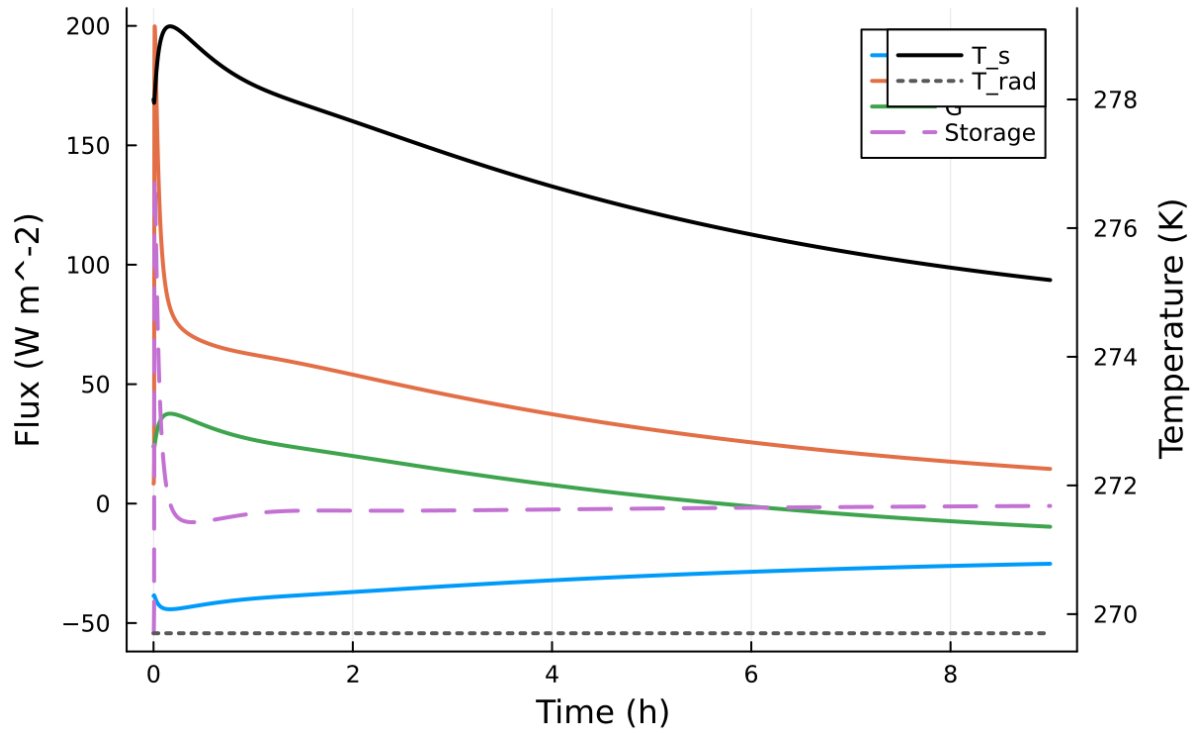


Figure 45: Surface energy budget components (R_n , H , G , storage)

6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

17.5 Included Plots

Figure 6: Regularized Slow-Manifold Phase Portrait

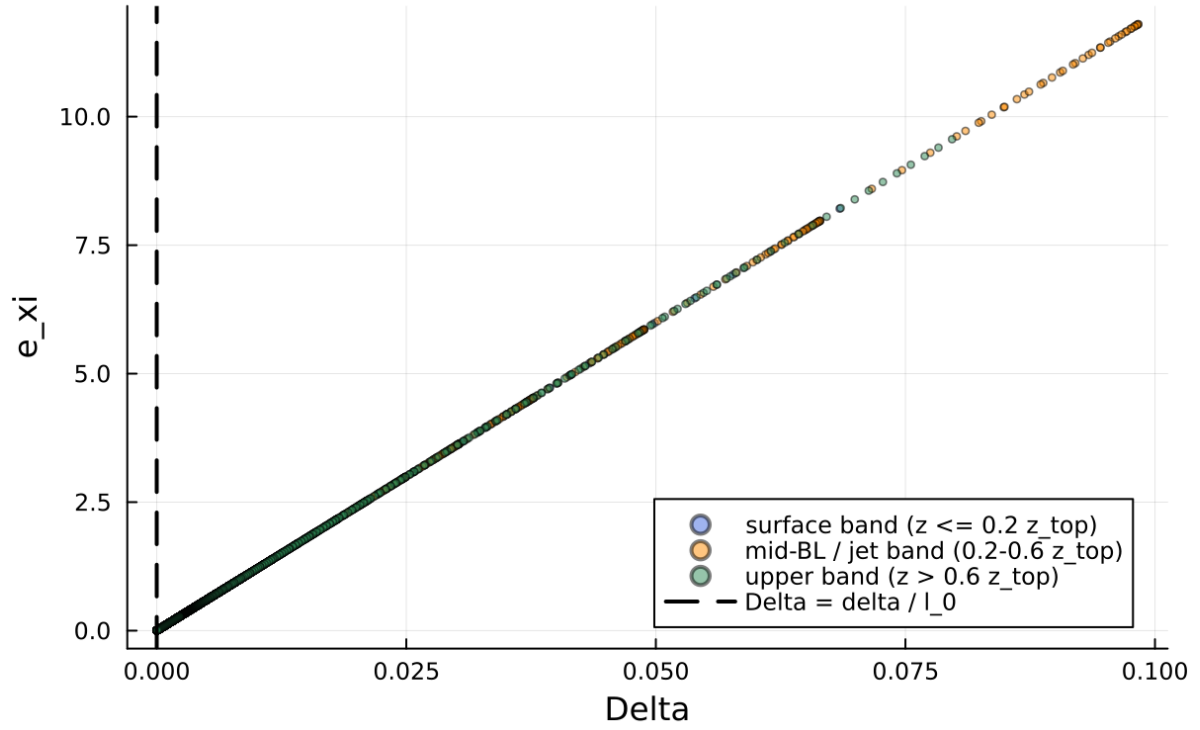


Figure 46: Phase portrait on regularized manifold (Δ vs e_{ξ})

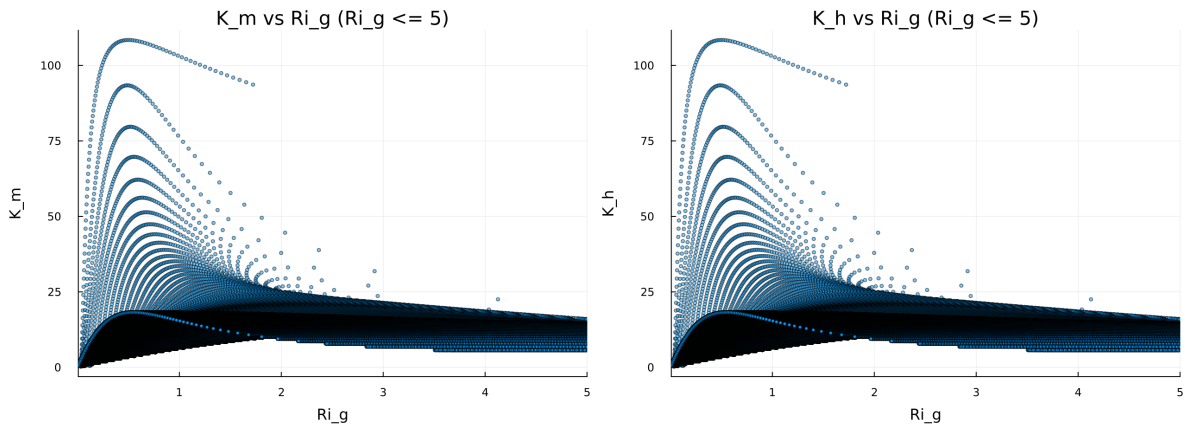


Figure 47: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

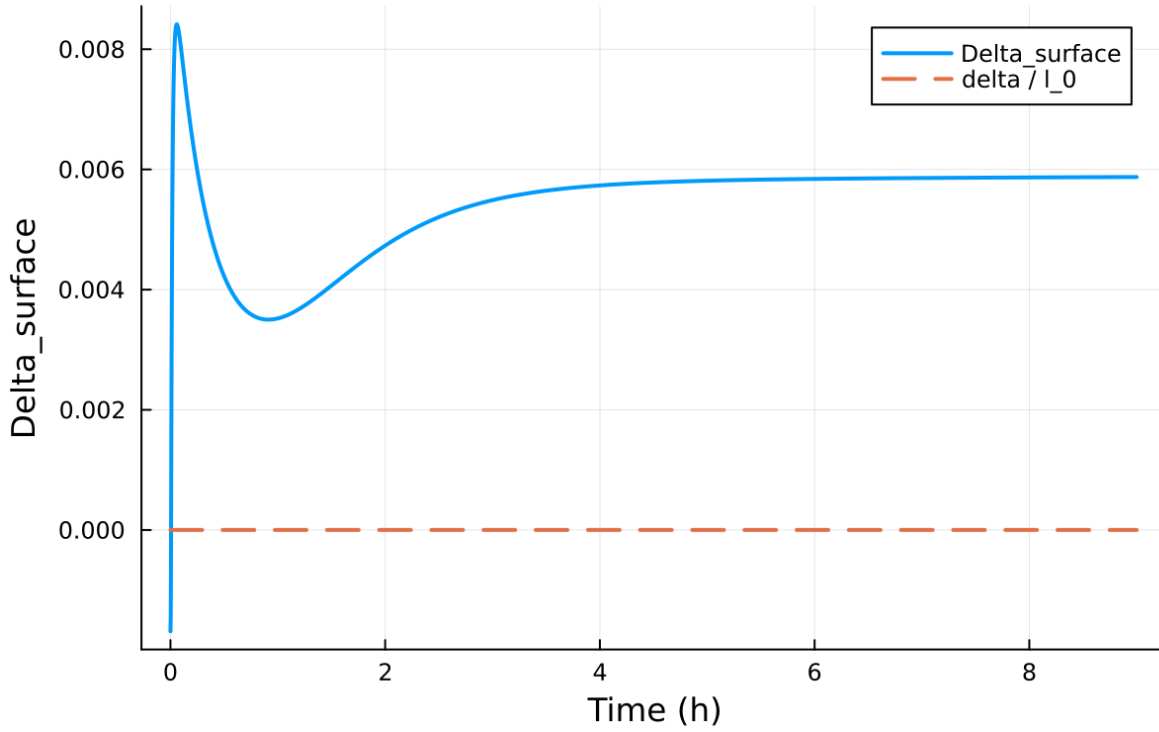


Figure 48: Fold proximity diagnostics versus time

18 Case 9: gabls1 150x2m 18h report

19 SCM Case Summary: gabls1

This section was generated automatically from the SCM diagnostics artifact `results/11j_regime/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver{Val{:forward}}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

19.1 Core Run Metadata

- Case identifier: `gabls1`
- Output directory: `results/11j_regime`
- Number of saved time samples: 2161
- Number of profile snapshots: 37
- Solver accepted/rejected steps: 491 / 2
- RHS evaluations: 226369
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

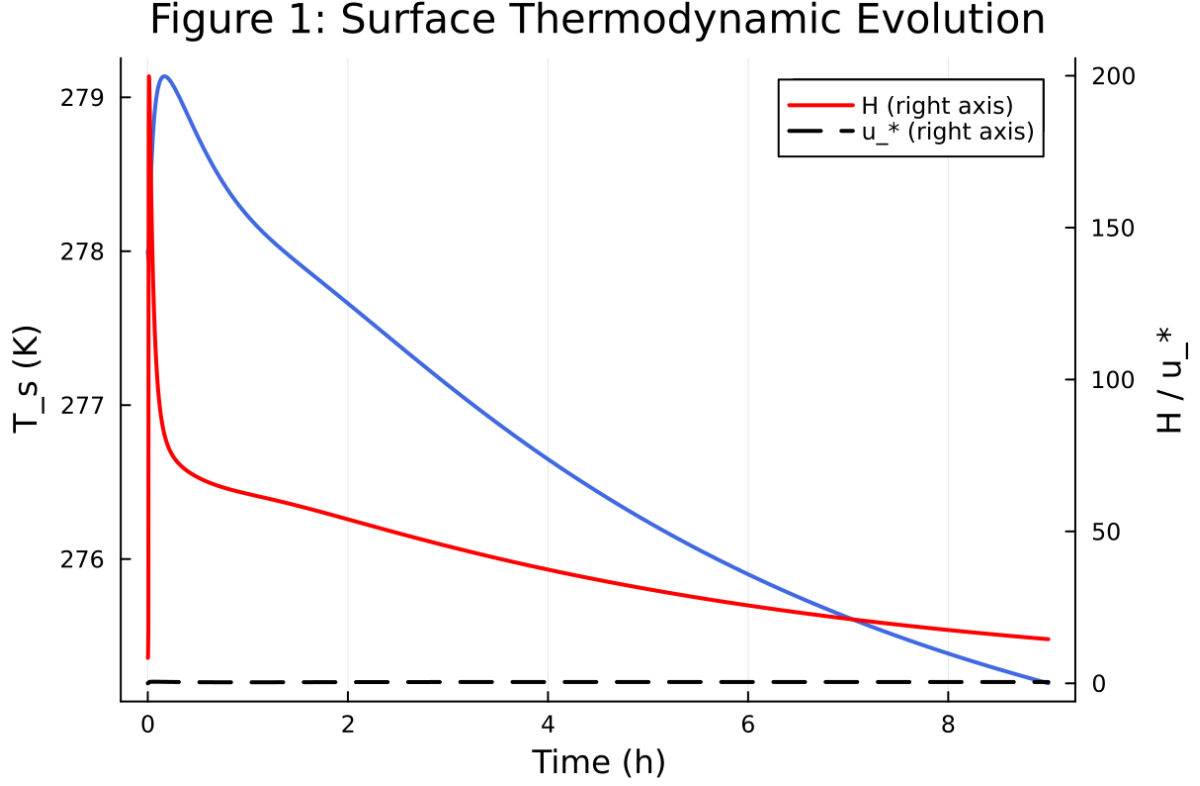


Figure 49: Time series of T_s , sensible heat flux, and friction velocity

19.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.103721, 79.407743] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.004032, 104.669656]$

19.3 Generated Artifacts

- Time-series CSV: `results/llj_regime/time_series.csv`
- Diagnostic summary JSON: `results/llj_regime/summary.json`
- Payload JLD2: `results/llj_regime/payload.jld2`

19.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

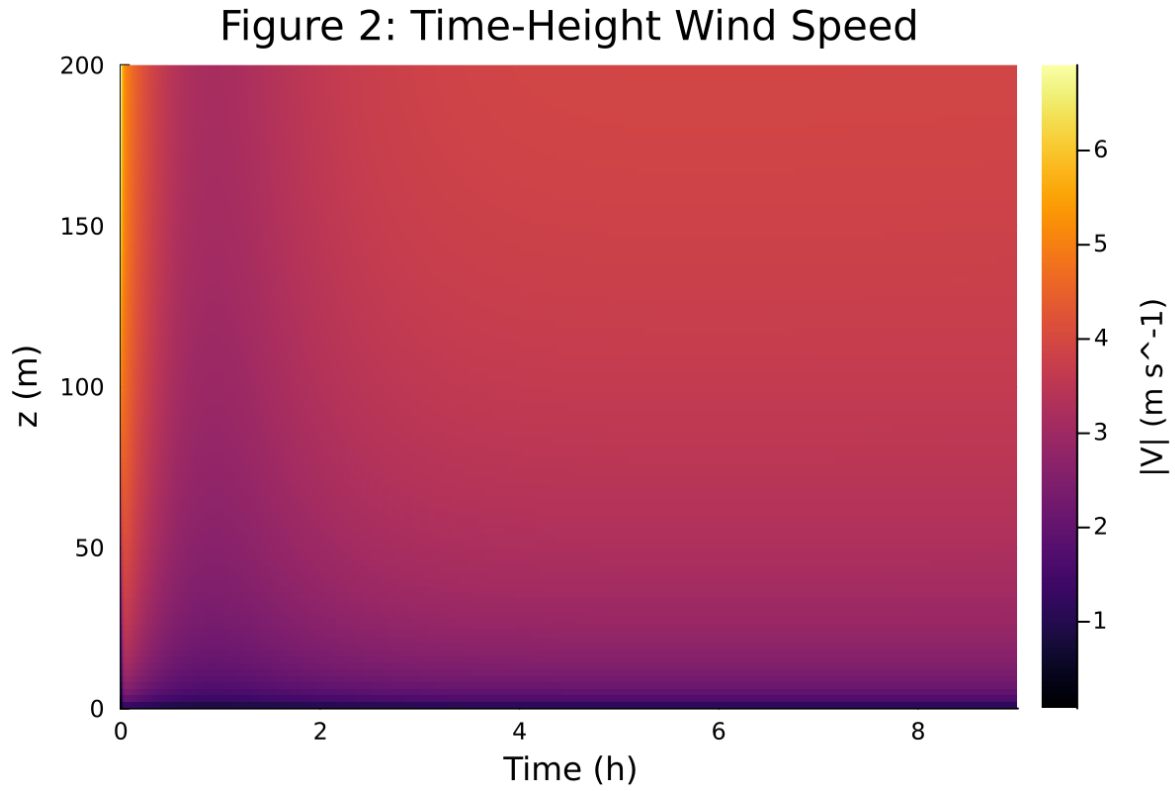


Figure 50: Time-height contour of wind speed with LLJ evolution

6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

19.5 Included Plots

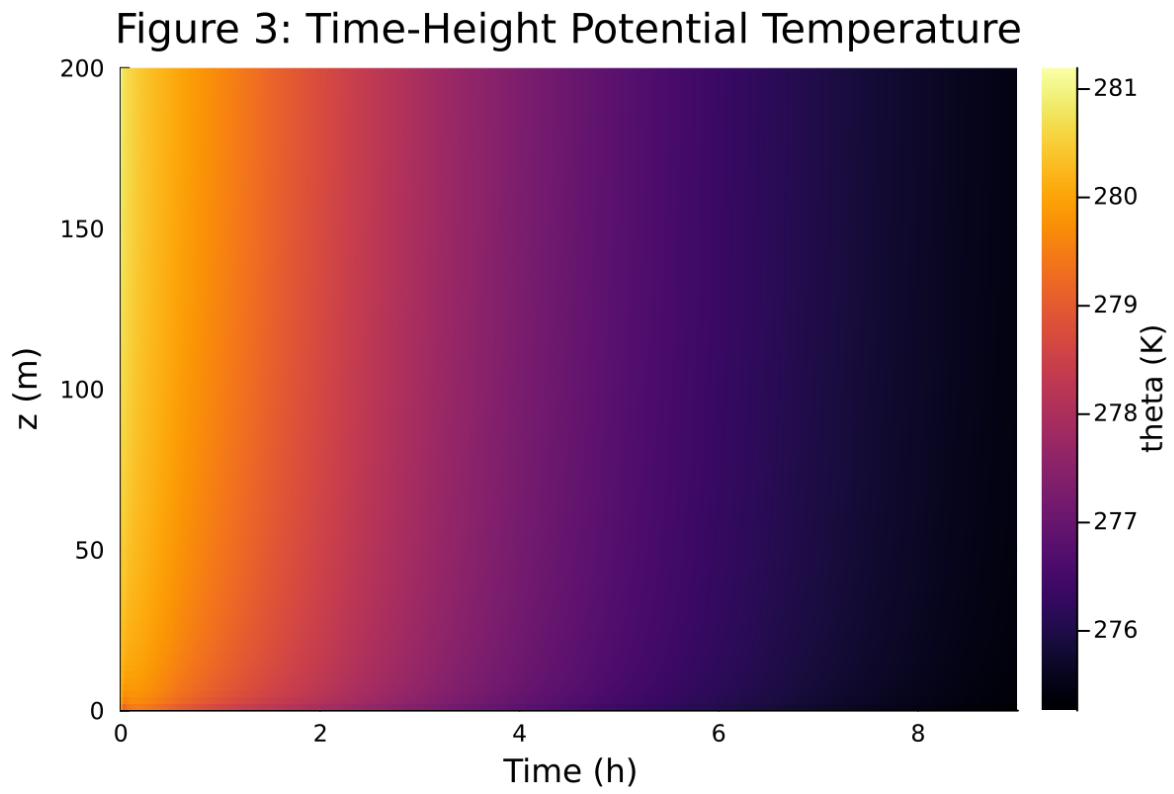


Figure 51: Time-height contour of potential temperature and inversion growth

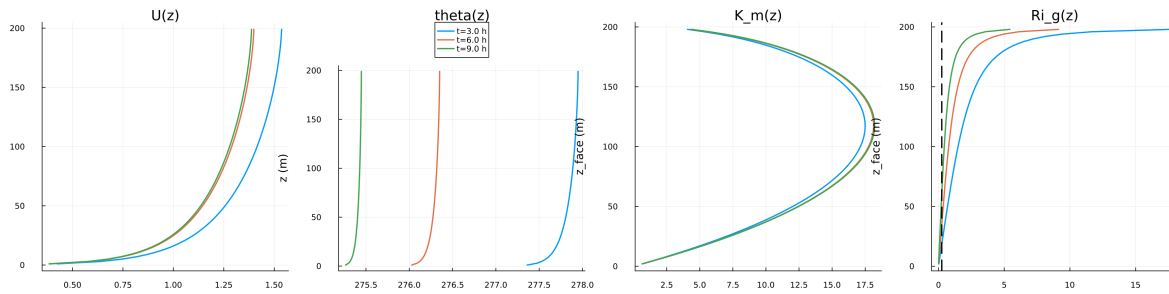


Figure 52: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

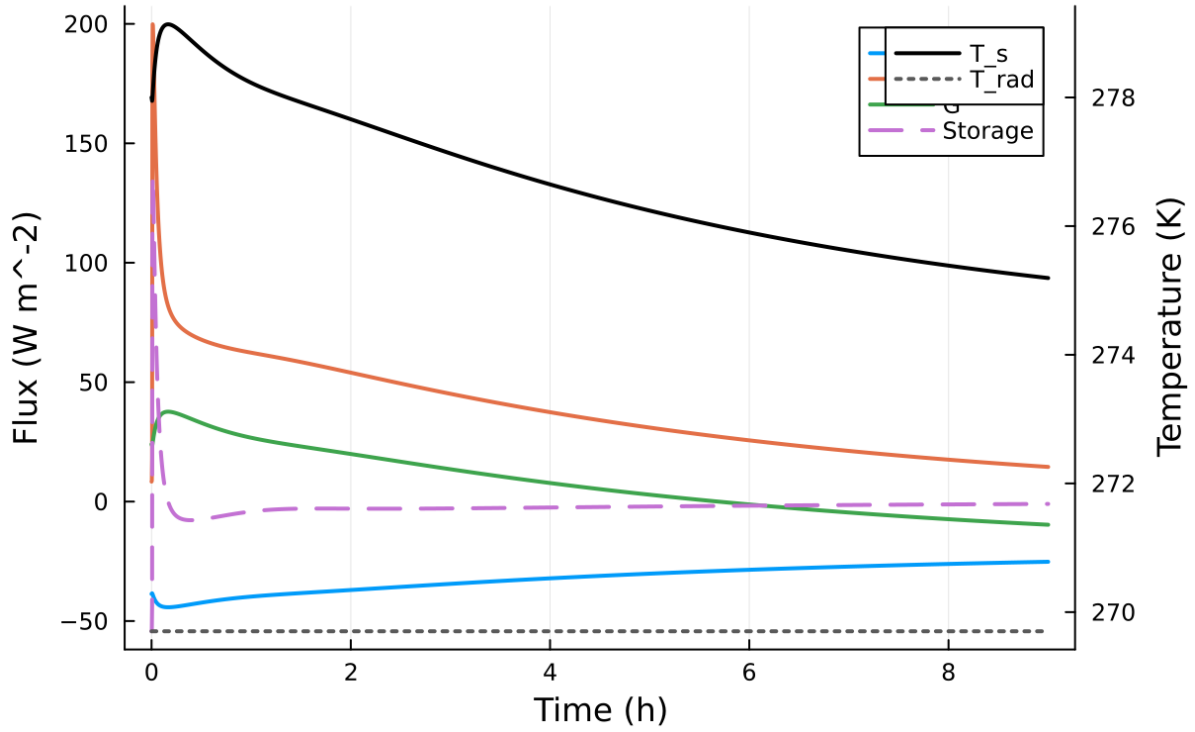


Figure 53: Surface energy budget components (R_n , H , G , storage)

20 Case 10: idealized sbl 40x4m 0h report

21 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver{Val{:forward}}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.trivial_limiter!)}` and returned `Success`.

21.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 60 / 3
- RHS evaluations: 7886
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

Figure 6: Regularized Slow-Manifold Phase Portrait

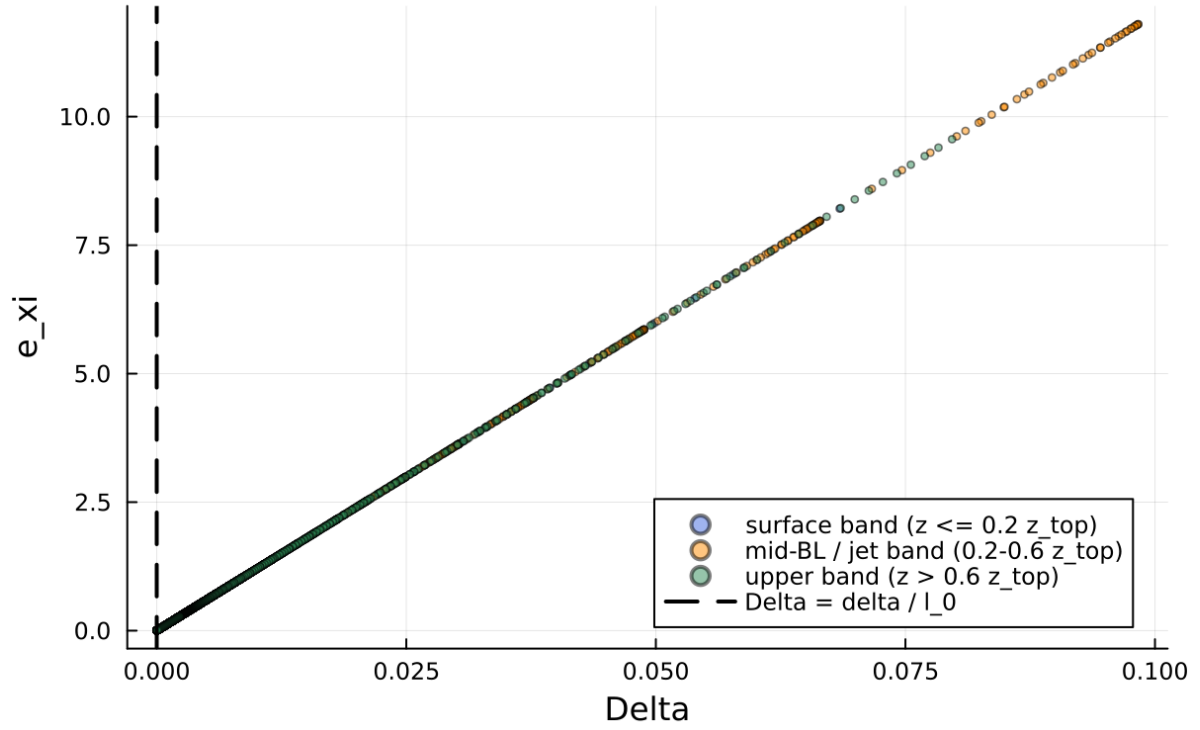


Figure 54: Phase portrait on regularized manifold (Δ vs e_{xi})

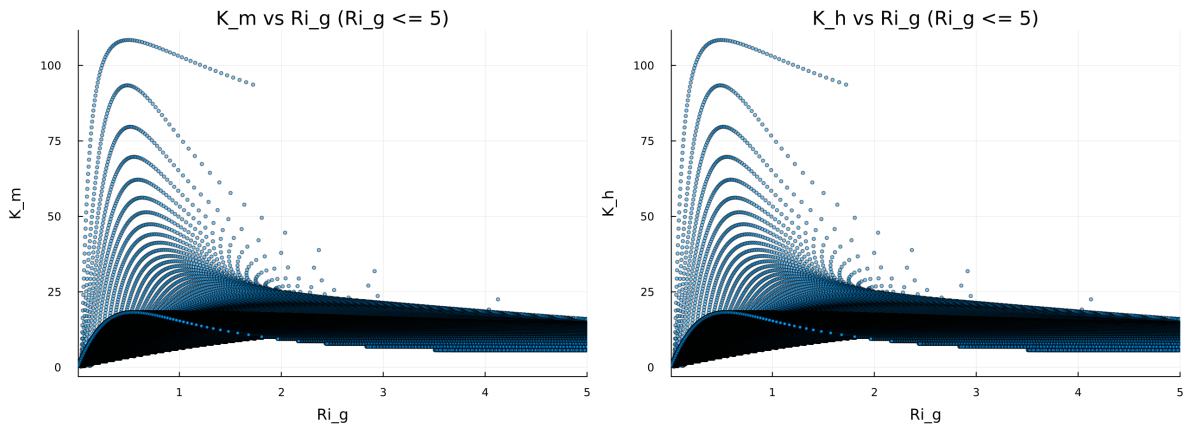


Figure 55: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

Figure 8: Fold Proximity Diagnostic

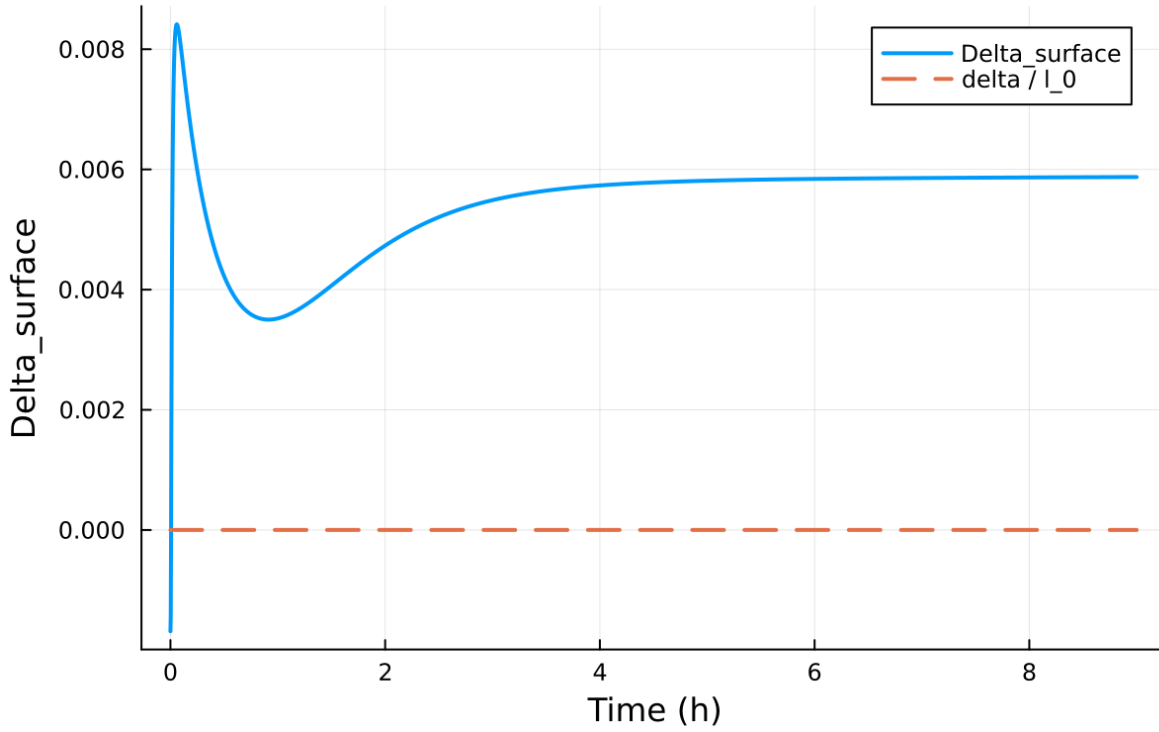


Figure 56: Fold proximity diagnostics versus time

21.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.719025, 108.431948] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.035577, 10.527066]$

21.3 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

21.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

Figure 1: Surface Thermodynamic Evolution

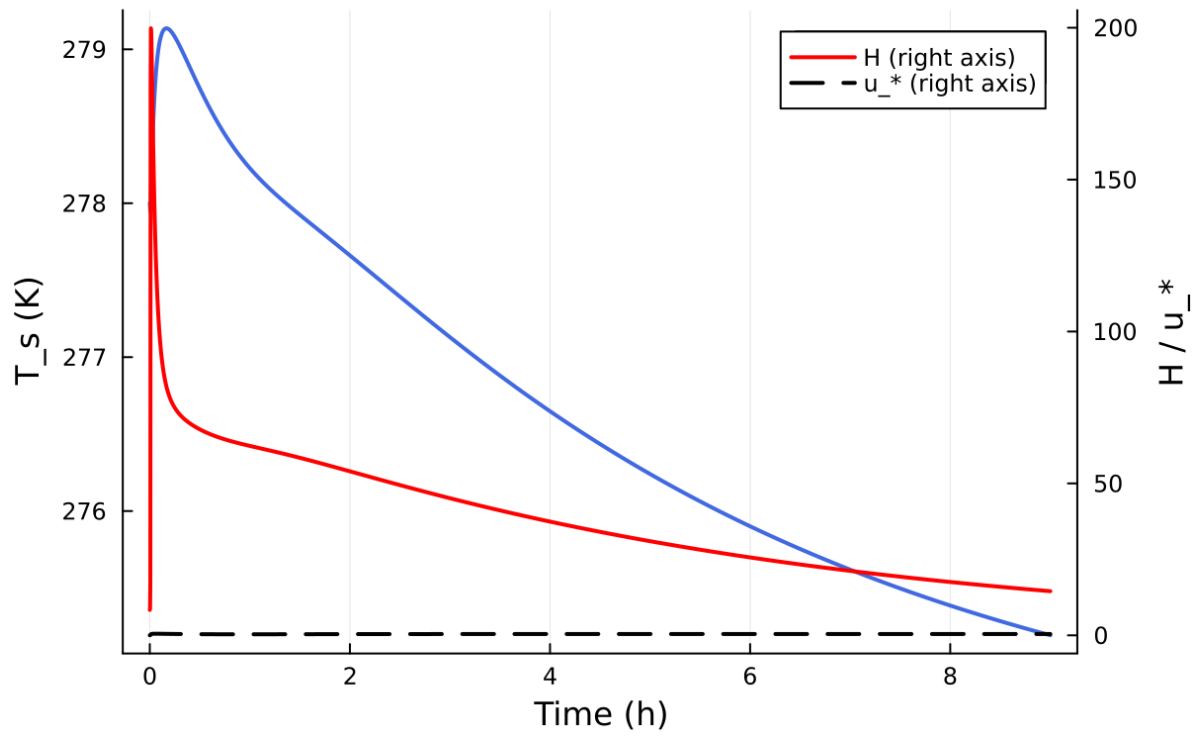


Figure 57: Time series of T_s , sensible heat flux, and friction velocity

6. Phase portrait on regularized manifold (Δ vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

21.5 Included Plots

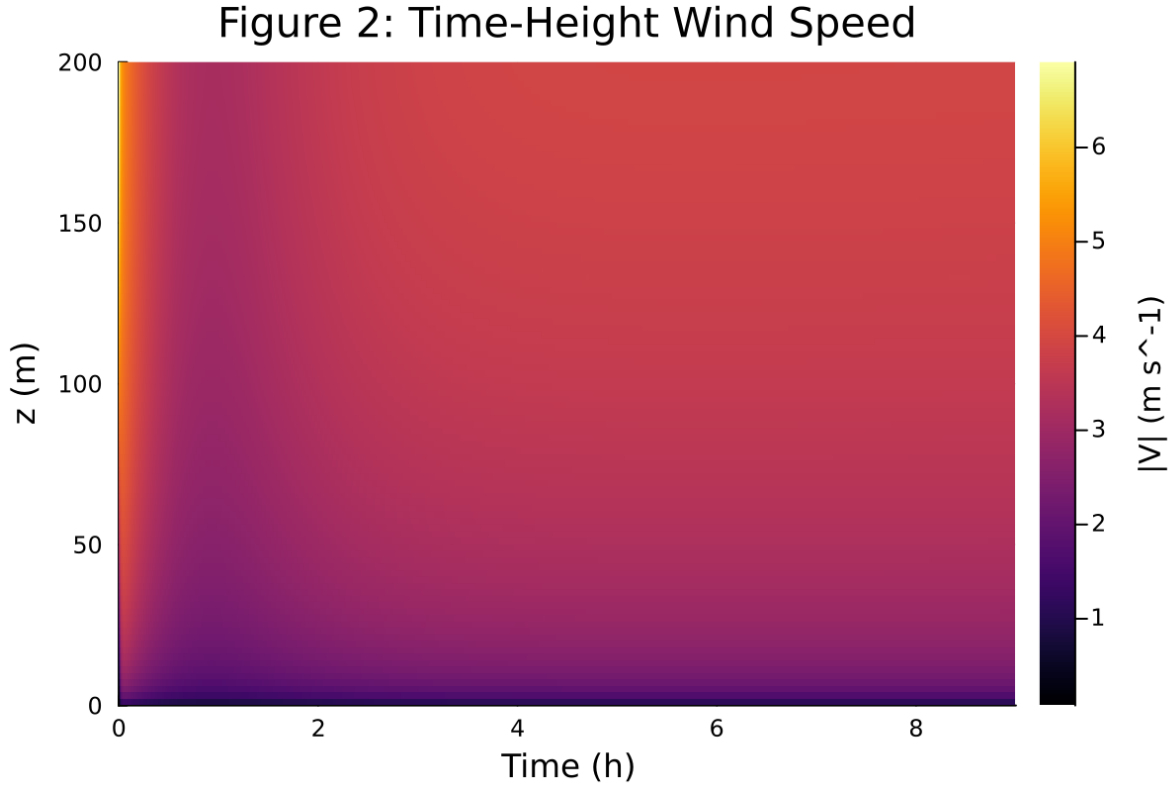


Figure 58: Time-height contour of wind speed with LLJ evolution

22 Case 11: scm verify 40x4m 0h report

23 SCM Case Summary: idealized_sbl

This section was generated automatically from the SCM diagnostics artifact `results/scm_verify/summary.json`. The simulation completed with `OrdinaryDiffEqRosenbrock.Rodas5P{0, ADTypes.AutoFiniteDiff{Val{:forward}, Val{:forward}, Val{:hcentral}}, Nothing, Nothing, Int64}, Nothing, typeof(OrdinaryDiffEqCore.DEFAULT_PREsolver{Val{:forward}}()), true, nothing, typeof(OrdinaryDiffEqCore.trivial_limiter!), typeof(OrdinaryDiffEqCore.tolerances)}` and returned `Success`.

23.1 Core Run Metadata

- Case identifier: `idealized_sbl`
- Output directory: `results/scm_verify`
- Number of saved time samples: 13
- Number of profile snapshots: 1
- Solver accepted/rejected steps: 60 / 3
- RHS evaluations: 7886
- Solver tolerances: $\text{abstol} = 1.000 \times 10^{-8}$, $\text{reitol} = 1.000 \times 10^{-6}$

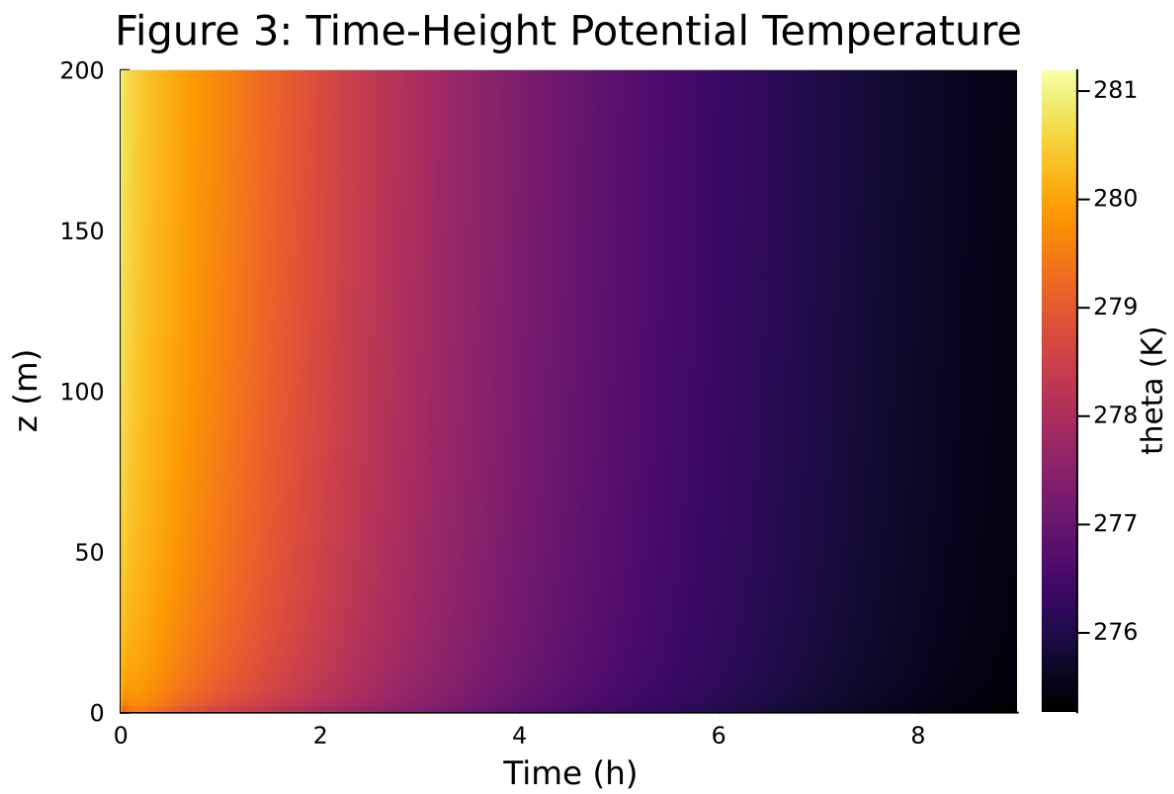


Figure 59: Time-height contour of potential temperature and inversion growth

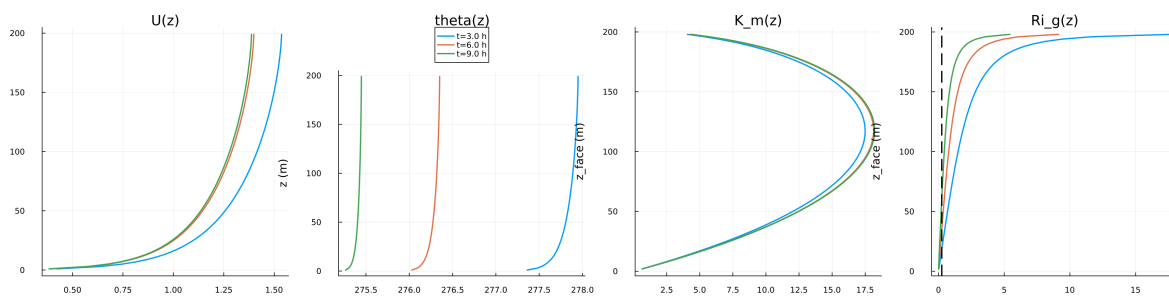


Figure 60: Vertical profiles of U, theta, and Km at representative times

Figure 5: Surface Energy Budget

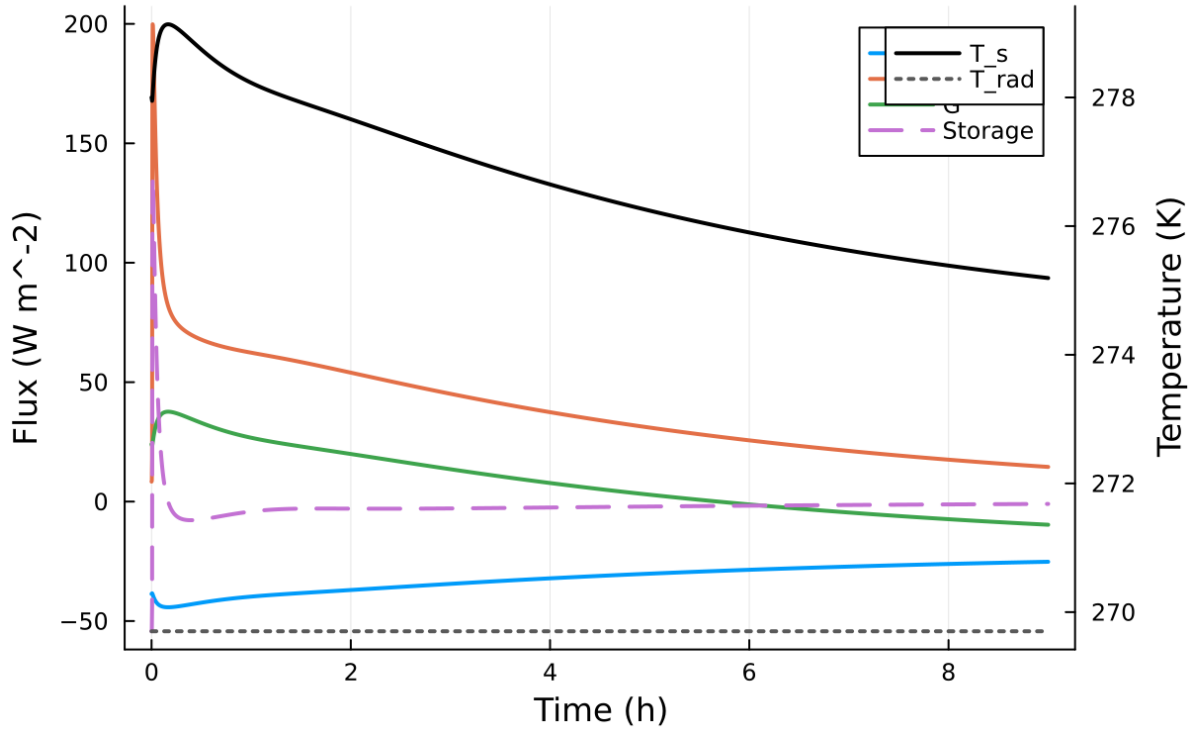


Figure 61: Surface energy budget components (R_n , H , G , storage)

23.2 Numerical Verification

- Maximum surface energy closure error: 0 W m^{-2}
- Fold-proximity fraction: 0.000000%
- Diffusivity bounds: $K_m \in [0.719025, 108.431948] \text{ m}^2 \text{ s}^{-1}$
- Richardson number bounds: $Ri \in [0.035577, 10.527066]$

23.3 Generated Artifacts

- Time-series CSV: `results/scm_verify/time_series.csv`
- Diagnostic summary JSON: `results/scm_verify/summary.json`
- Payload JLD2: `results/scm_verify/payload.jld2`

23.4 Figure Manifest

1. Time series of T_s , sensible heat flux, and friction velocity
2. Time-height contour of wind speed with LLJ evolution
3. Time-height contour of potential temperature and inversion growth
4. Vertical profiles of U , θ , and K_m at representative times
5. Surface energy budget components (R_n , H , G , storage)

Figure 6: Regularized Slow-Manifold Phase Portrait

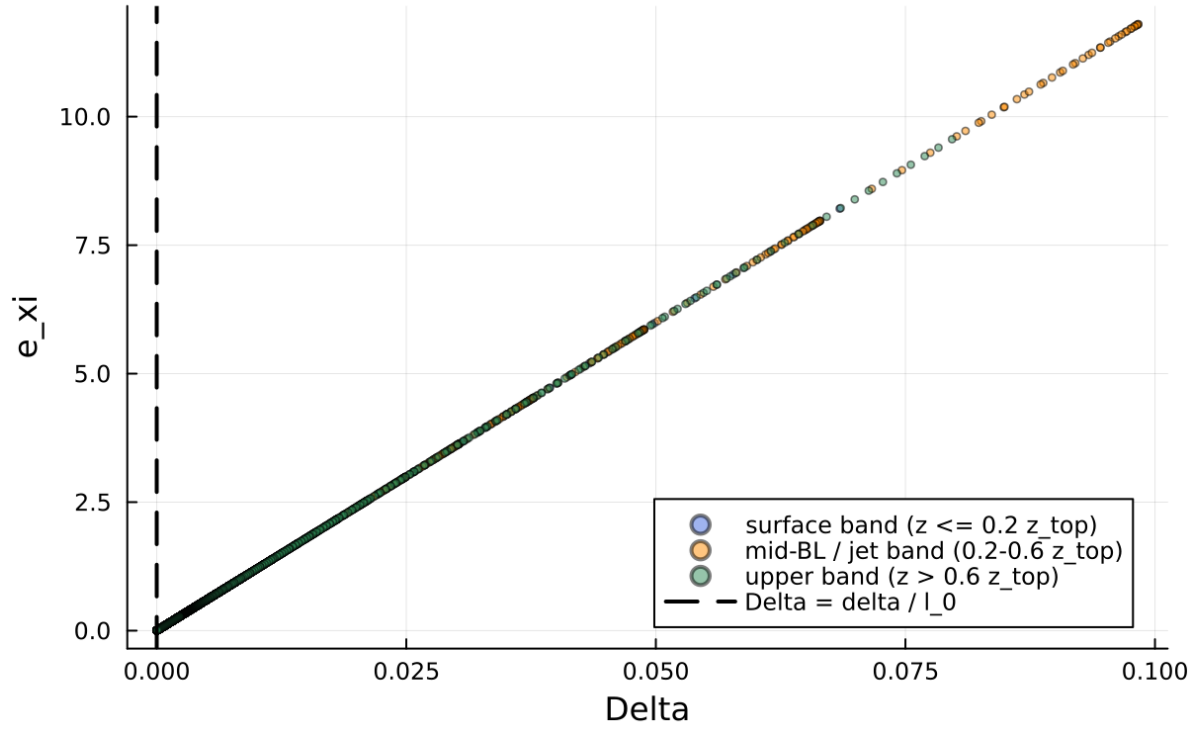


Figure 62: Phase portrait on regularized manifold (Δ vs e_{xi})

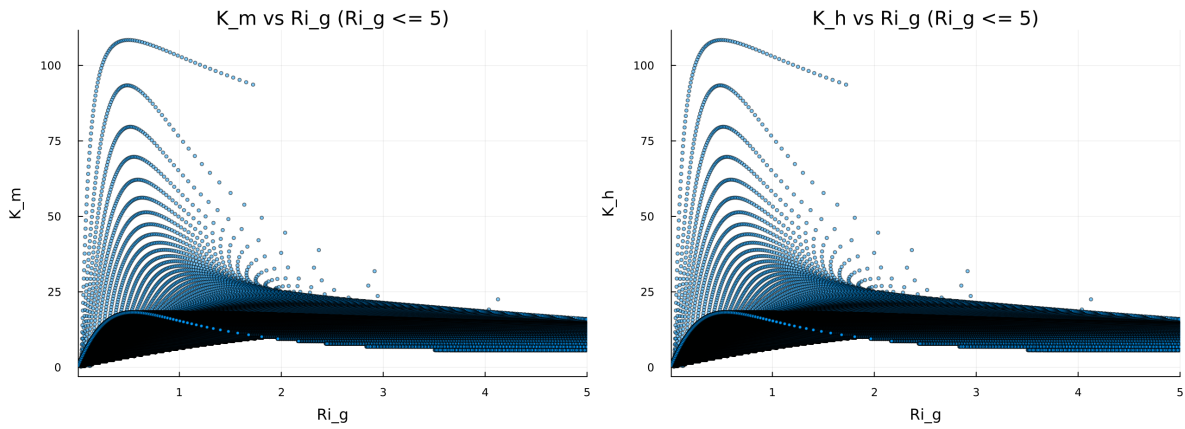


Figure 63: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

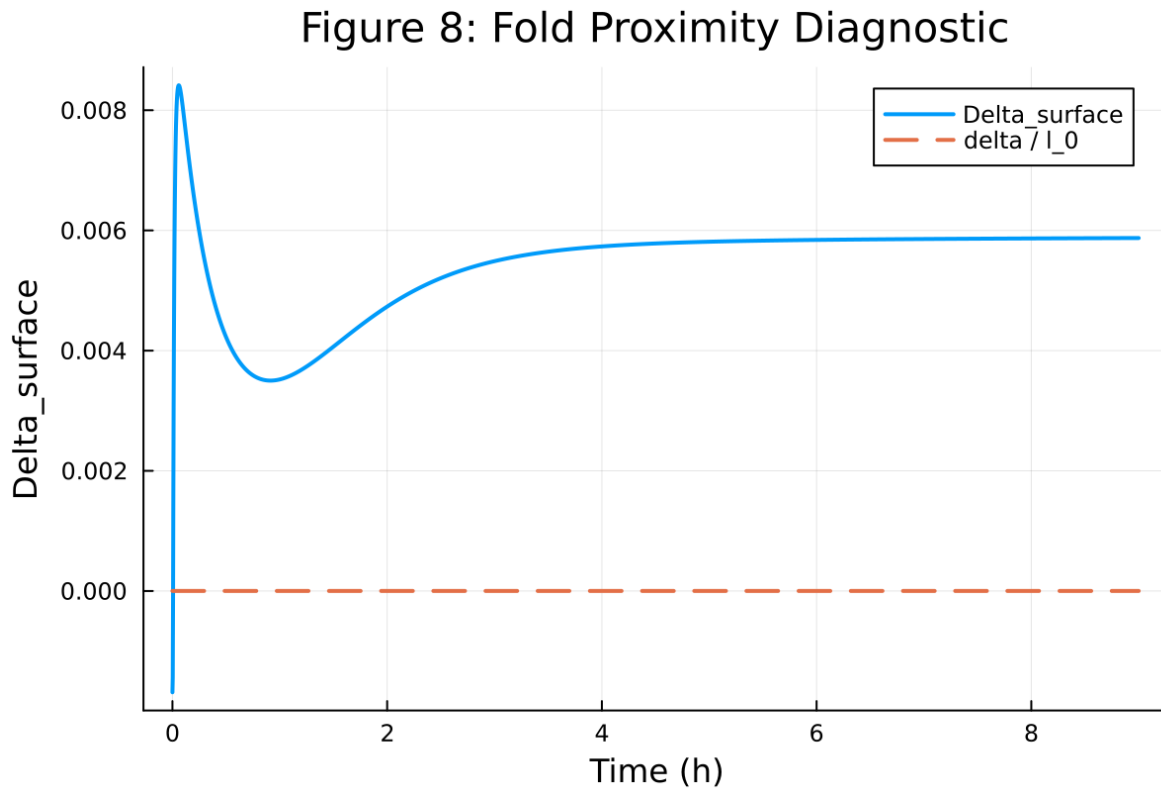


Figure 64: Fold proximity diagnostics versus time

6. Phase portrait on regularized manifold (Delta vs e_{xi})
7. Eddy diffusivity response (K_m , K_h) versus stability/Richardson number
8. Fold proximity diagnostics versus time

23.5 Included Plots

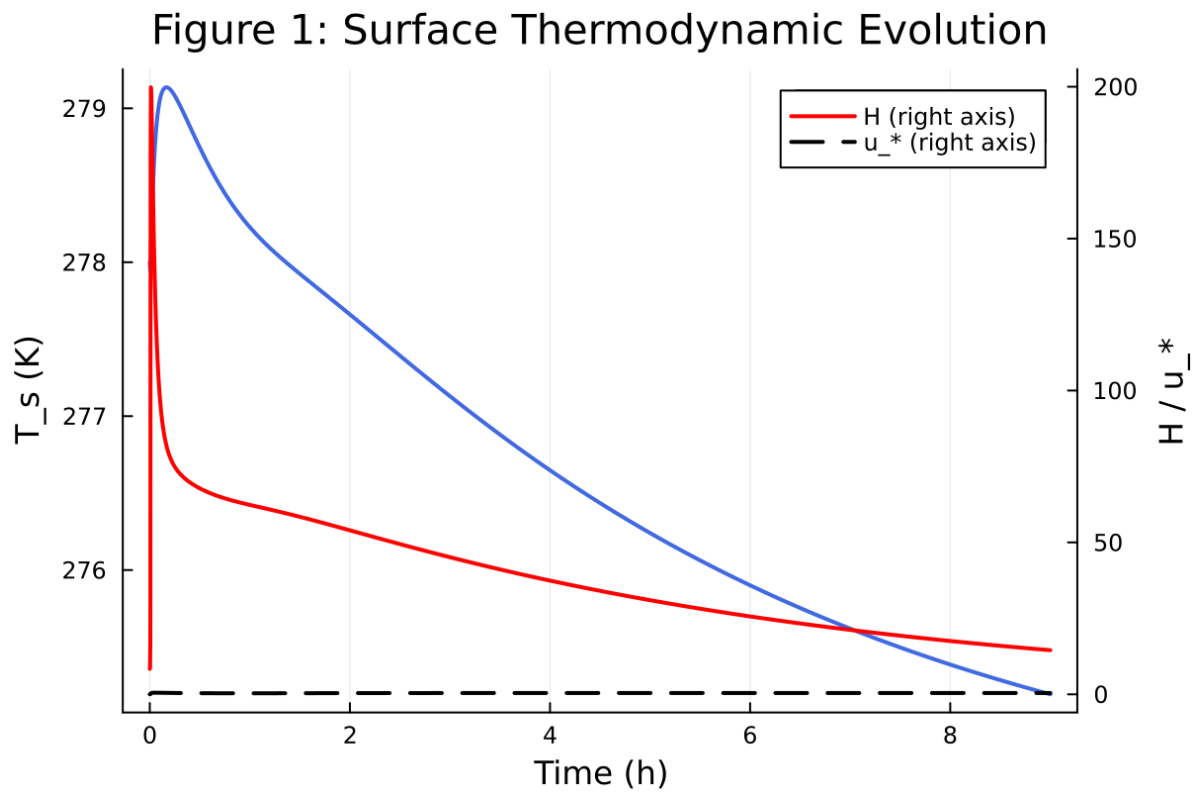


Figure 65: Time series of T_s , sensible heat flux, and friction velocity

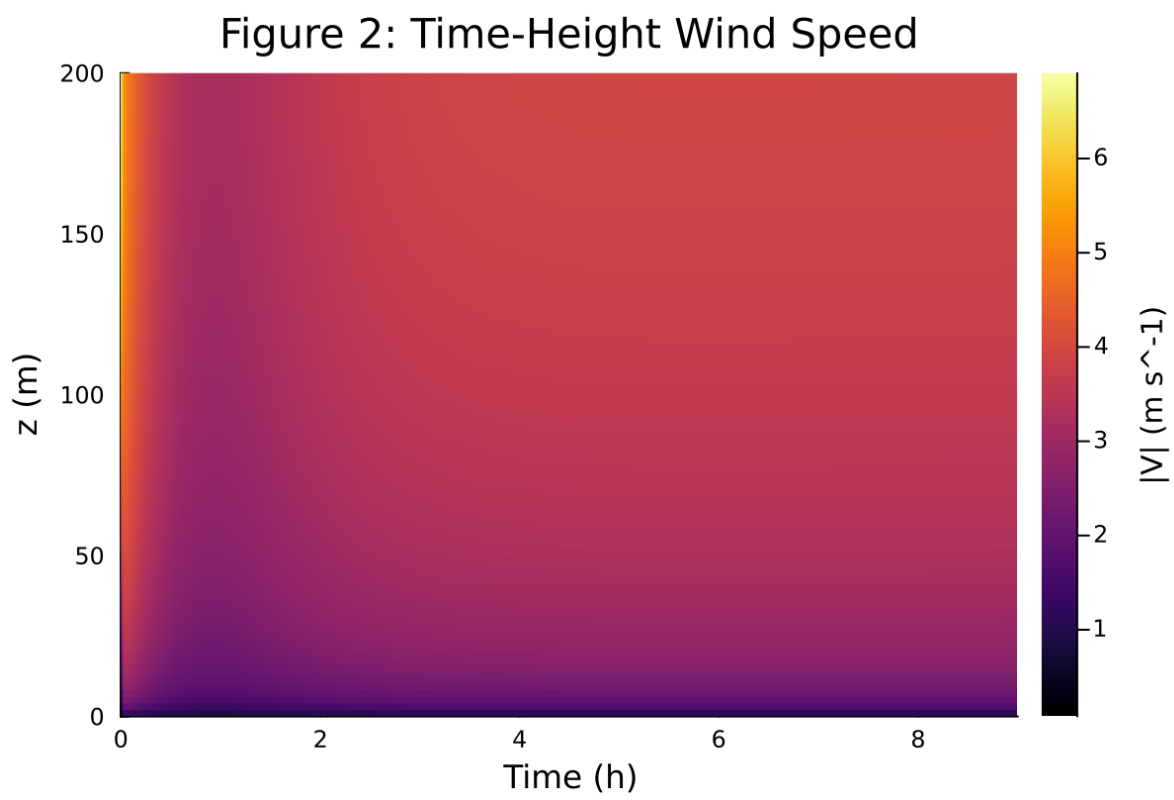


Figure 66: Time-height contour of wind speed with LLJ evolution

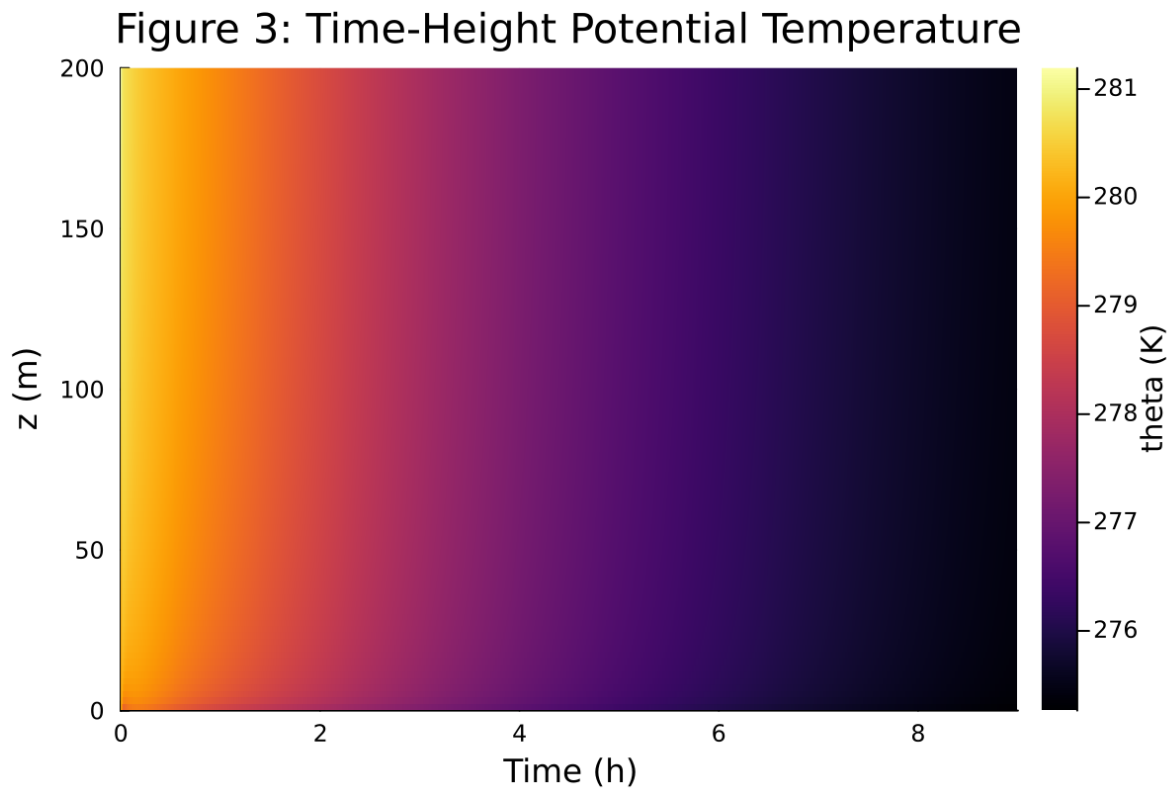


Figure 67: Time-height contour of potential temperature and inversion growth

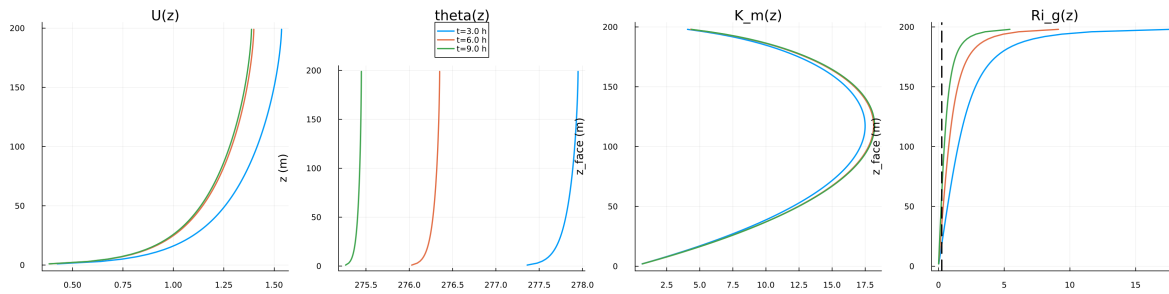


Figure 68: Vertical profiles of U, theta, and Km at representative times

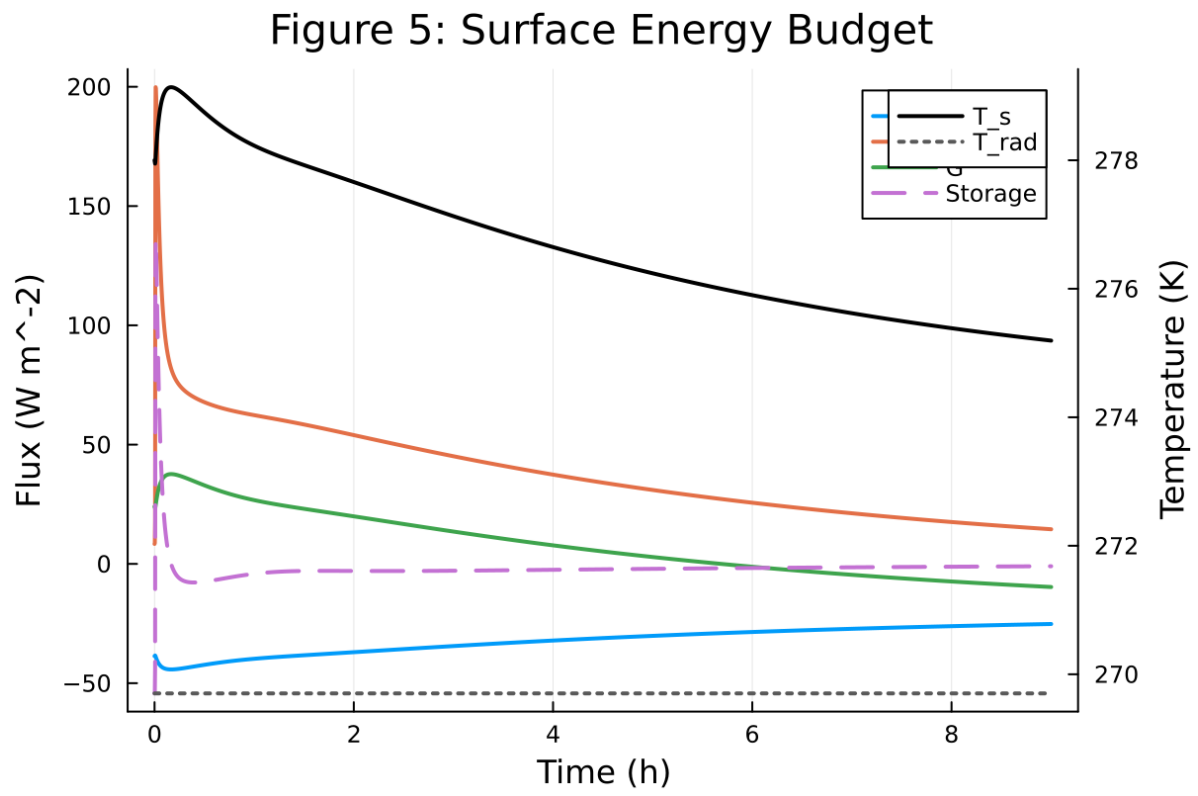


Figure 69: Surface energy budget components (R_n , H , G , storage)

Figure 6: Regularized Slow-Manifold Phase Portrait

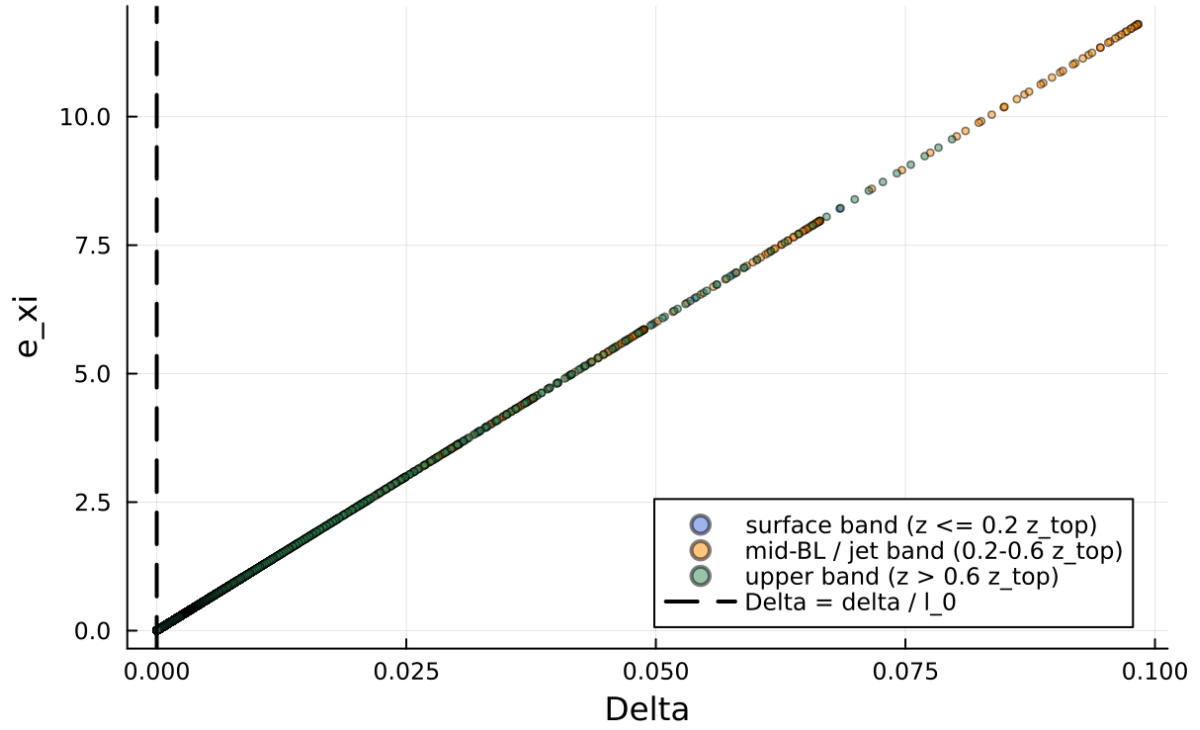


Figure 70: Phase portrait on regularized manifold (Δ vs e_{xi})

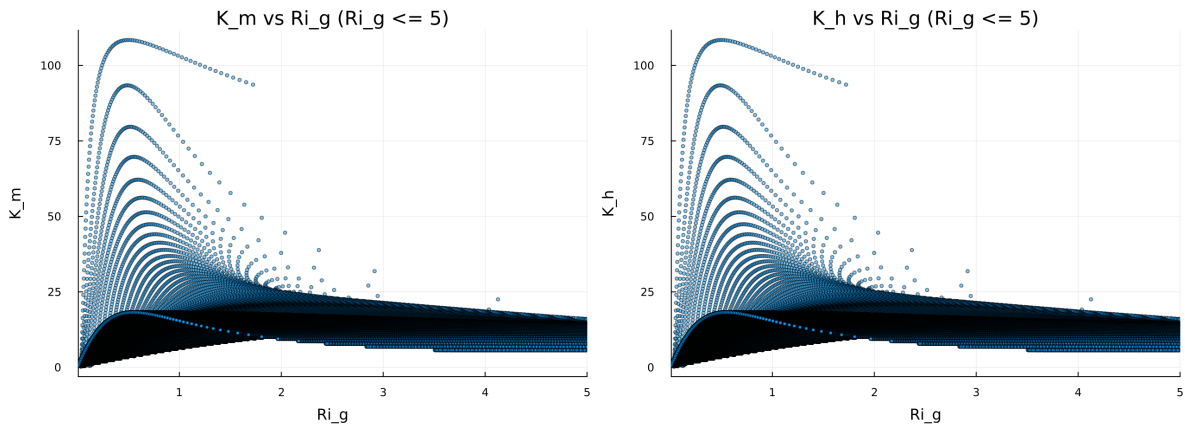


Figure 71: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

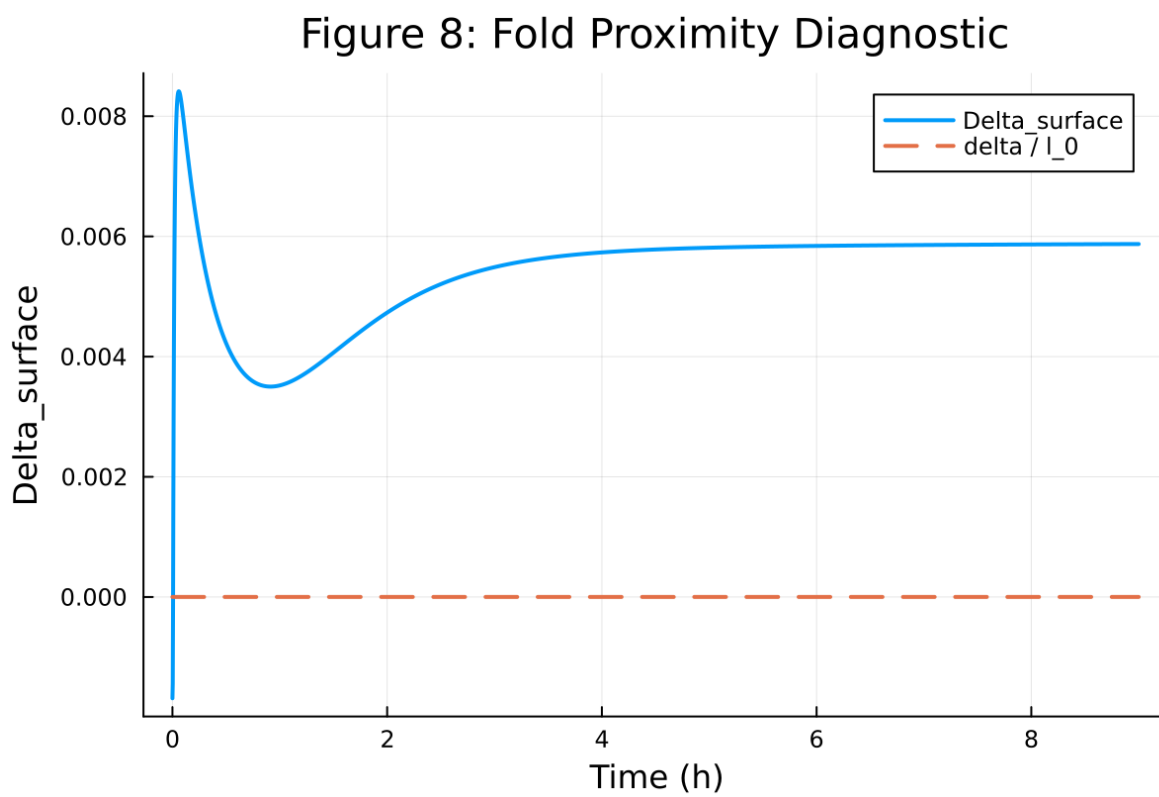


Figure 72: Fold proximity diagnostics versus time

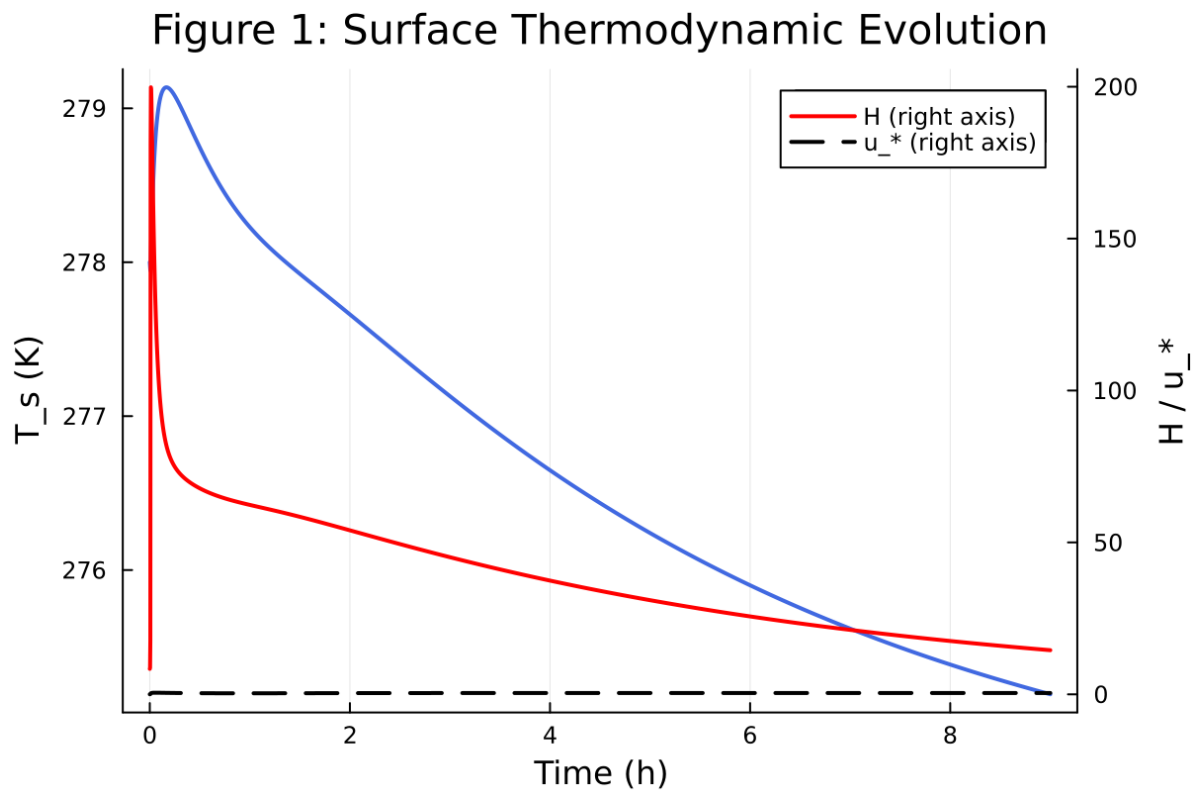


Figure 73: Time series of T_s , sensible heat flux, and friction velocity

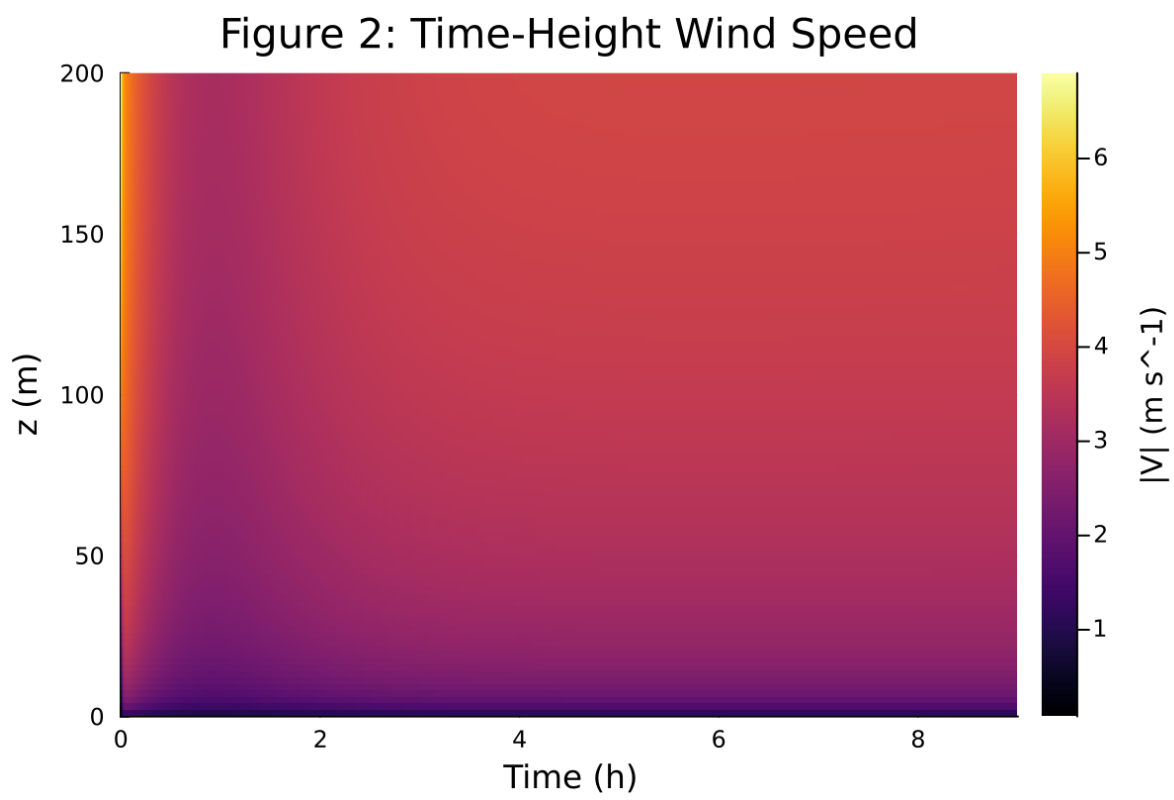


Figure 74: Time-height contour of wind speed with LLJ evolution

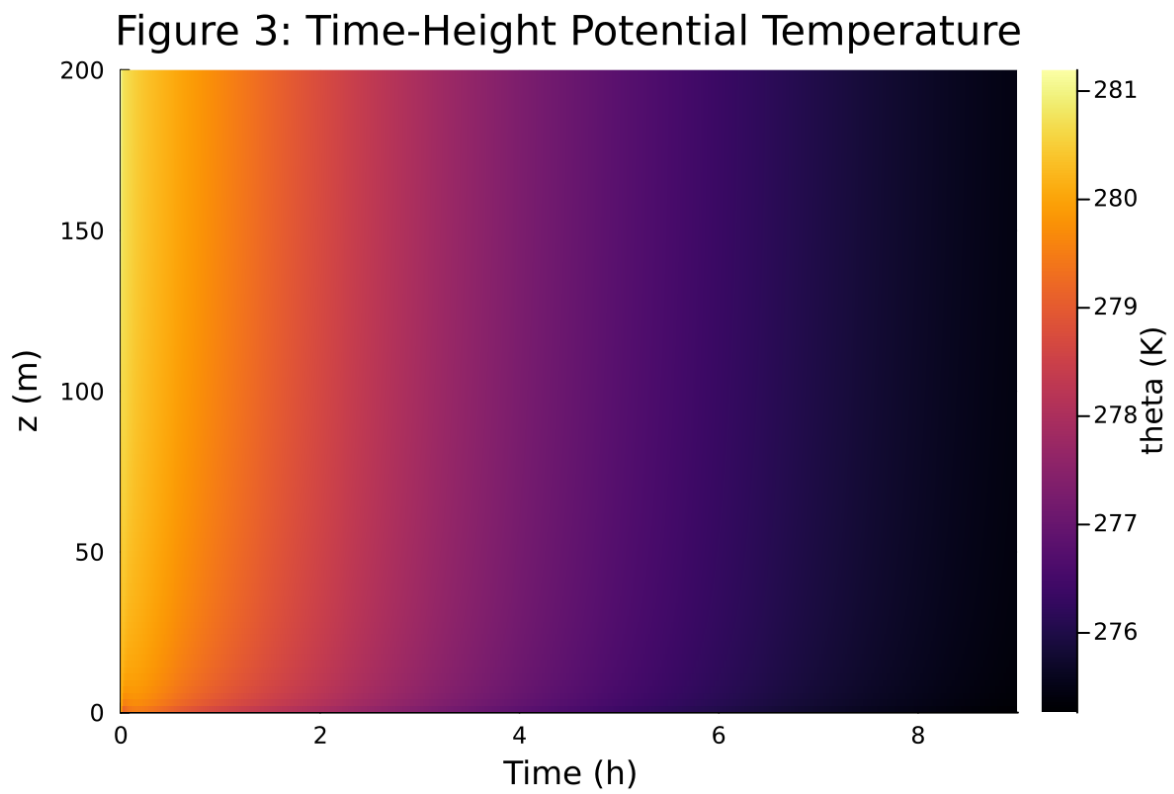


Figure 75: Time-height contour of potential temperature and inversion growth

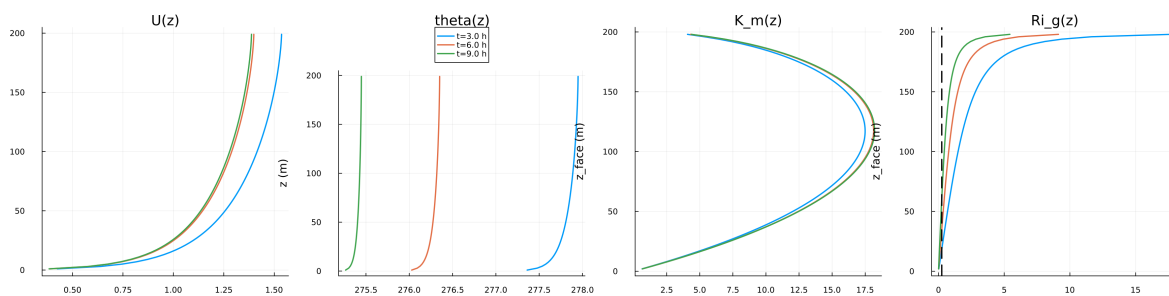


Figure 76: Vertical profiles of U, theta, and Km at representative times

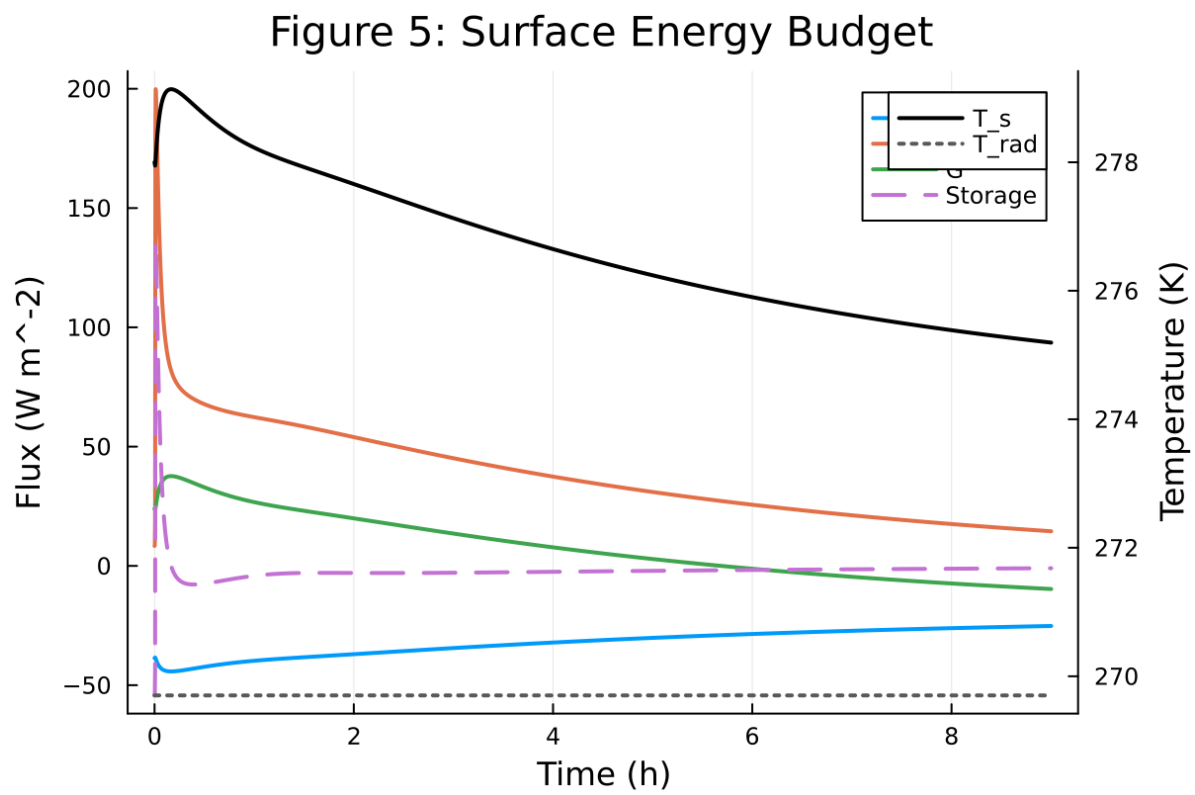


Figure 77: Surface energy budget components (R_n , H , G , storage)

Figure 6: Regularized Slow-Manifold Phase Portrait

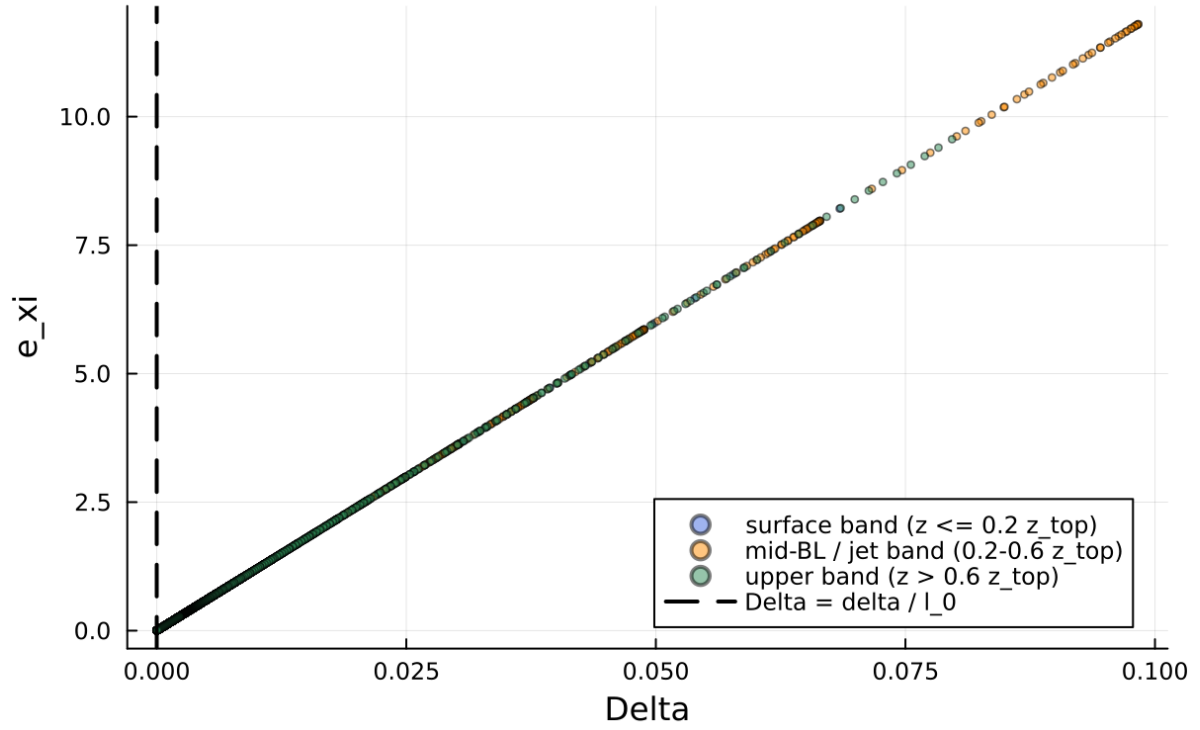


Figure 78: Phase portrait on regularized manifold (Delta vs e_xi)

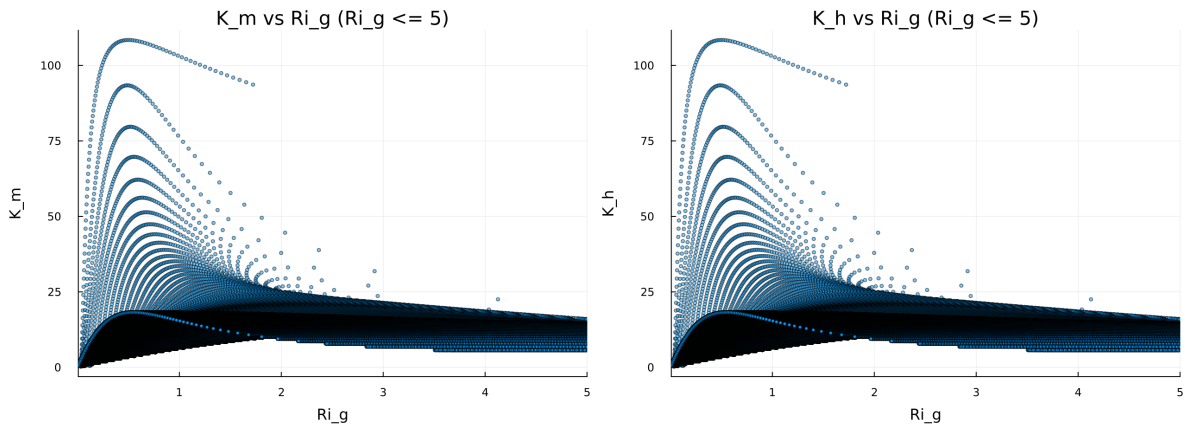


Figure 79: Eddy diffusivity response (K_m , K_h) versus stability/Richardson number

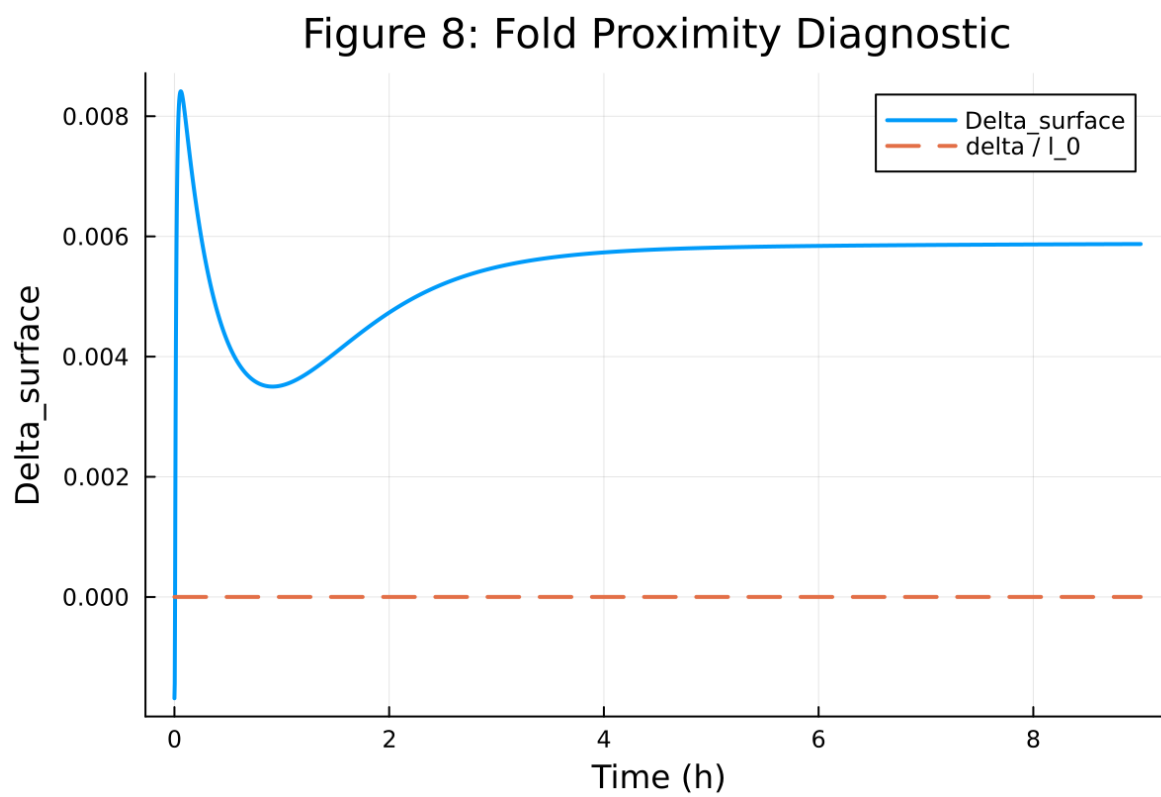


Figure 80: Fold proximity diagnostics versus time